MSIA: A Multiscale Interactive Attention Network Assisted by Self-Supervised Contrastive Learning for Hyperspectral Image Classification

Gu Gong, Jiahua Zhang, Xiaopeng Wang, Xiaodi Shang, *Member, IEEE*, Zhicheng Pan, Jingyuan Wang, and Xianwei Huo

***Abstract*—Hyperspectral image (HSI) classification is a crucial research area in hyperspectral remote sensing, aiming to identify the land cover type of each pixel in remote sensing images using computer analysis. This task faces two major challenges: 1) patch-based models typically operate at a fixed scale, limiting their ability to capture cross-scale interactions and often blurring the contribution of the central pixel due to interference from neighboring pixels; and 2) labeled samples in hyperspectral datasets are extremely limited, making supervised training prone to overfitting. To address these issues, we propose a multiscale interactive attention network (MSIA) that explicitly divides each patch into three concentric regions-center, near, and relevant-to overcome the single-scale limitation. The center region is processed through multiple Transformer Encoders to enhance target-pixel perception, while the near and relevant regions interact via a central attention block (CAB), enabling effective localglobal feature exchange. To mitigate the challenge of label scarcity, we further introduce a self-supervised contrastive pretraining strategy, which leverages abundant unlabeled samples and fine-tunes the network with only a small set of labeled data. Experiments on five benchmark datasets demonstrate that the proposed method achieves state-of-the-art classification performance under limited supervision.**

***Index Terms*—Contrastive learning, deep learning, global multiscale attention, hyperspectral image (HSI), multiscale interactive, multiscale interactive attention network (MSIA), self-attention.**

I. Introduction

With the development of imaging spectrometers and advancements in space technology, remote sensing has entered the hyperspectral stage. Unlike traditional RGB images, hyperspectral images (HSIs) contain rich spatial and spectral information, with spectral bands spanning from visible light to infrared, covering tens or even hundreds of narrow bands. These images are widely used in mineral exploration [1], environmental monitoring [2], agricultural detection [3], medical diagnosis [4], and military target detection [5], with HSI classification becoming a core task in these applications.

Unlike traditional image classification, HSI classification faces the challenge of the "curse of dimensionality" due to its high spectral resolution. Without dimensionality reduction, directly training classifiers on high-dimensional raw data can result in poor performance [6]. Therefore, dimensionality reduction is a common preprocessing step. Principal component analysis (PCA) [7] and linear discriminant analysis (LDA) [8] are the most classical unsupervised and supervised feature extraction methods. Early studies often relied on expert knowledge for feature extraction, such as wavelet transforms, spectral indices, and spectral feature fitting, which were then input into traditional machine learning models for classification [9], [10], [11]. While these methods promoted the progress of HSI classification, manually crafted features often lack generalization and are limited in complex scenarios.

In contrast to expert-driven shallow features, deep learning methods automatically learn discriminative representations from raw data with minimal human intervention, attracting significant attention [12], [13], [14], [15], [16], [17]. Early HSI classification methods like deep belief networks (DBNs) [18], stacked autoencoders (SAEs) [19], and recurrent neural networks (RNNs) [20] primarily modeled spectral information. However, despite the high spectral resolution of HSI, the loss of spatial structural information severely impacts classification performance due to the "same object with different spectra" and "different objects with the same spectrum" phenomena. Convolutional neural networks (CNNs) subsequently became mainstream in HSI classification by effectively extracting both spatial and spectral features [21], [22], [23]. Early works, such as Hu et al [24], used 1-D CNNs for spectral feature extraction. Subsequently, 2-D and 3-D CNNs were developed to fuse spatial and spectral features, filling research gaps in spectral–spatial fusion for HSI classification [25]. Beyond pure CNNs, researchers have combined them with other

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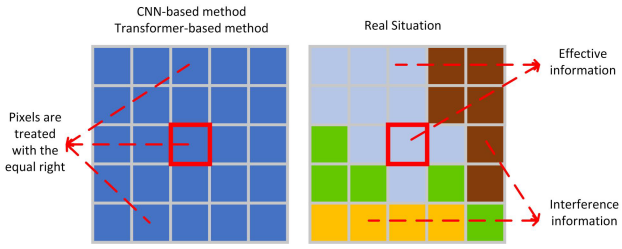

Fig. 1. Equal treatment of pixels in CNNs and Transformers versus real-world spatial variability.

techniques: Chen et al. [26] integrated Gabor filters with CNNs for preprocessing. Zhong et al. [27] incorporated residual learning modules into 3D-CNNs, increasing network depth while ensuring stable convergence. Roy et al. [28] designed a hybrid 3-D–2-D CNN architecture, improving accuracy while reducing parameters.

Recently, Transformer-based models, known for their self-attention mechanism, have emerged as cutting-edge methods in HSI classification [29], [30], [31], [32]. Hong et al. [33] proposed SpectralFormer, leveraging neighboring spectral bands to learn sequential features. Peng et al. [34] designed a cross-attention mechanism capable of capturing spatial features and establishing deep dependencies between spectral sequences. Although Transformers excel at capturing long-range dependencies, their ability to model local context is limited. Consequently, researchers often combine them with CNNs to compensate for this shortcoming [35], [36]. Xu et al. [37] proposed a bilateral classification network, first extracting multiscale spatial features using a CNN, then fusing these convolutional features with the global attention of a Transformer to deeply integrate the long-range dependency features of HSI and LiDAR data. Sun et al. [38] proposed the SSFTT network, which uses convolutional operations in the front end to extract local features and increase dimensionality, followed by a Transformer for classification.

Despite the promising performance of these deep learning methods, several challenges remain. First, the geometric constraints of input data lead to a single perspective for the model. HSI data often exhibits complex spatial patterns and diverse distribution characteristics, where spatial information at multiple scales is crucial for accurate classification. Larger scales capture regional patterns, while smaller scales help discern fine-grained structures. However, current CNN or Transformer-based methods mostly use fixed-size patches as inputs, limiting the model's ability to capture multiscale spatial features and reducing information exchange across different spatial positions.

Second, the importance of the central pixel in classification is often underappreciated. In practice, neighboring pixels with similar spectral characteristics can provide useful spatial context, whereas those with significantly different properties may introduce noise. As illustrated in Fig. 1, conventional CNNs, due to the weight-sharing mechanism of convolutional filters, implicitly treat pixels within the receptive field in a uniform manner, without explicitly reinforcing the discriminative primacy of the central pixel. Transformer models, on the other hand, can in principle highlight informative pixels via attention scores. However, by modeling pairwise dependencies among all pixels in a patch without structural constraints, they introduce substantial computational overhead and may propagate redundant or interfering information.

To address these issues, this article proposes a multiscale interactive attention network (MSIA). The network divides the input patch into three subregions: center region, near region, and relevant region, and processes each region differently based on its importance in classification. The center region undergoes rearrangement and is processed through multiple Transformer encoders to enhance the central pixel's contribution to the classification decision. The near and relevant regions execute a central attention block (CAB), where the near region is mapped to a lower dimension and combined with the relevant region for query-key-value (QKV) computation. This reduces redundant information exchange and enhances computational efficiency. The features from the center region and the CABs are then concatenated and passed through a multilayer perceptron (MLP) for classification.

In addition to the architectural challenge, a second major obstacle lies in the scarcity of labeled hyperspectral samples. Ground-truth annotation is labor-intensive and costly, leading to datasets with only a limited number of labels. As a result, purely supervised training is prone to overfitting and poor generalization. Recently, self-supervised contrastive learning has shown great promise in computer vision by exploiting abundant unlabeled data to learn discriminative representations.

To overcome this problem, we integrate a self-supervised contrastive pretraining strategy into MSIA. The model is first pretrained on large amounts of unlabeled hyperspectral data to capture transferable spectral–spatial representations, and then fine-tuned with a small labeled set. This integration enhances the robustness of MSIA under limited supervision, enabling it to simultaneously address both the structural challenge of multiscale feature interaction and the practical challenge of label scarcity. Taken together, these considerations highlight a critical research gap: although prior studies have advanced either architectural innovations or label-efficient learning strategies, few have addressed both aspects in a unified framework. Our work fills this gap by combining explicit multiscale interaction modeling with self-supervised contrastive learning.

The main contributions of this article can be summarized as follows.

1) We design an MSIA that partitions input patches into hierarchical regions and applies differentiated strategies to capture spatial information at multiple scales.
2) The model explicitly emphasizes the central pixel, reinforcing key discriminative features for improved classification.
3) A CAB is introduced to enhance cross-scale feature interaction while reducing irrelevant information and computation overhead.
4) We integrate self-supervised contrastive pretraining to leverage unlabeled data and alleviate label scarcity in HSI classification.

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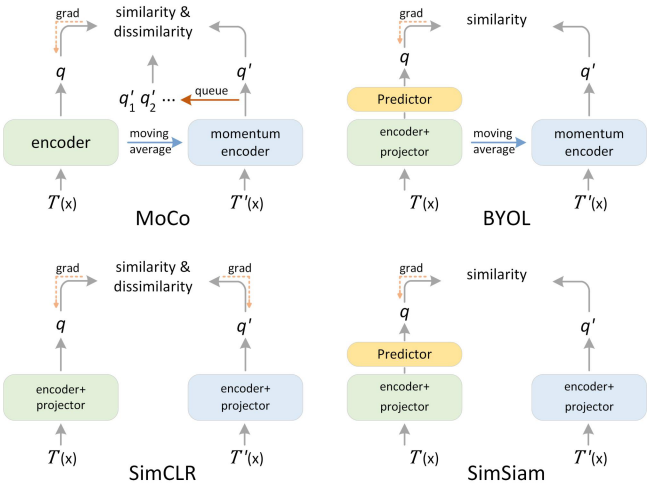

Fig. 2. Structure of different contrastive learning methods.

II. Related Works

A. Comparative Learning

Traditional deep learning models often struggle to learn generalizable features from limited labeled data, especially for high-dimensional or complex categories, leading to overfitting or underfitting. One advanced training approach is self-supervised contrastive learning, which uses a Siamese network to simulate supervised tasks by contrasting different views (e.g., augmentations) of the same image, guiding the model to learn discriminative feature representations without the need for manual annotations [39], [40], [41]. The trained model is then fine-tuned on downstream tasks with a small amount of labeled data.

Contrastive learning can be categorized based on the use of negative samples. As illustrated in Fig. 2, momentum contrast (MoCo) [42] maintains a dynamic queue of historical samples as negatives and uses a momentum-updated key encoder to ensure feature consistency. Meanwhile, SimCLR [43] employs a simpler architecture, generating positive and negative pairs directly within the same batch. A key contribution of SimCLR was the introduction of a nonlinear projection head (e.g., a two-layer MLP) after the encoder, a simple modification that significantly boosted downstream task performance.

Although this contrastive learning strengthens feature representations by pulling positives together and pushing negatives apart, it requires numerous negative samples and a high computational cost. To alleviate this, BYOL [44] introduces a predictor after the projection head to predict the target encoder's output, removing the need for negatives. It also employs momentum updates in an asymmetric architecture to prevent collapse. SimSiam [45] further simplifies BYOL by discarding the momentum encoder and stopping the gradient flow of the target branch to achieve stability.

B. Contrastive Loss

The core of contrastive learning is to construct relationships between similar and dissimilar samples, thereby guiding the network to learn discriminative representations. A key to this process is the contrastive loss function. Early methods like SimCLR and MoCo adopted the InfoNCE loss, which aims to maximize the similarity between different views of the same sample (positive pairs) and minimize the similarity with other samples (negative pairs). Its formulation is as follows:

$$L = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(z_i, z_j)/\tau)}. \tag{1}$$

Here, Z_i, and Z_i^+ represent the two different augmentations of the same image, Z_j denotes negative samples, $\text{sim}(\cdot,\cdot)$ is the cosine similarity function, τ is the temperature coefficient, and N is the total number of samples, including positive samples. As contrastive learning evolved without the need for negative samples, the loss function was further simplified. Typically, negative cosine similarity is used to achieve the task, and the general form is

$$L(p, z) = -\frac{p^{\mathrm{T}} z}{\|p\| \cdot \|z\|} \tag{2}$$

where P and Z represent the output of the predictor and another view encoder, and $\|\cdot\|$ denotes the L2 norm used for normalization.

III. Methodology

This section details the proposed MSIA model architecture and its training method. Fig. 3 illustrates the overall framework of the supervised MSIA model. After PCA dimensionality reduction, the data is divided into regions. To ensure that the sampling windows at each scale are centered around the target pixel, we define the sizes of the center region, near region, and relevant region as $W \times W \times B, (2W+1) \times (2W+1) \times B$, and $(4W+3) \times (4W+3) \times B$, respectively, where W denotes the region size, and B is the number of principal components selected after PCA.

Subsequently, the rearranged and embedded center region is input into the Transformer, while the near region and relevant region jointly perform the CAB. Finally, the outputs from both paths are concatenated and fused through an MLP for class prediction.

A. Center Region

The center region plays a significant role in classification, especially in scenarios with highly similar land cover, where its local features effectively enhance interclass separability. To strengthen the feature representation of the center region, we use a Transformer encoder for feature extraction.

As shown in Fig. 4, to adapt to the Transformer's sequence input requirement, the data is first serialized. Since hyperspectral data is inherently 3-D, no additional mapping is introduced during serialization. The region is flattened into a sequence $C = [c_1, c_2, \ldots, c_t] \in R^{t \times B}$, where $t = W \times W$. A learnable global feature vector $c_{\text{class}} \in R^{1 \times B}$ is added at the beginning of the sequence to integrate global information, forming the input matrix $C' = [c_{\text{class}}, C] \in R^{U \times B}$, where $U = t + 1$. To preserve spatial position information, a learnable positional encoding vector $P_c = [\text{pos}_{\text{class}}, \text{pos}_1, \ldots, \text{pos}_t] \in R^{U \times B}$ is introduced. By adding this positional vector to the feature

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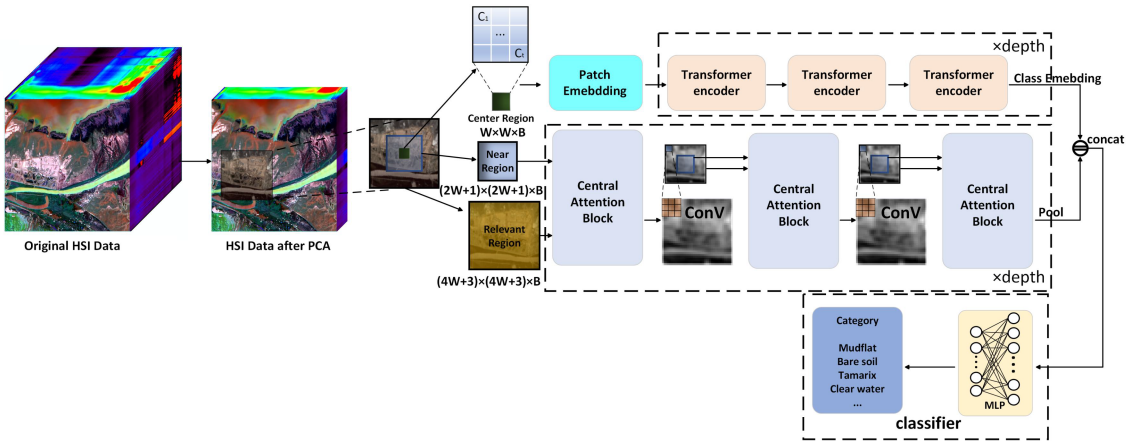

Fig. 3. Framework of the proposed method with supervised training.

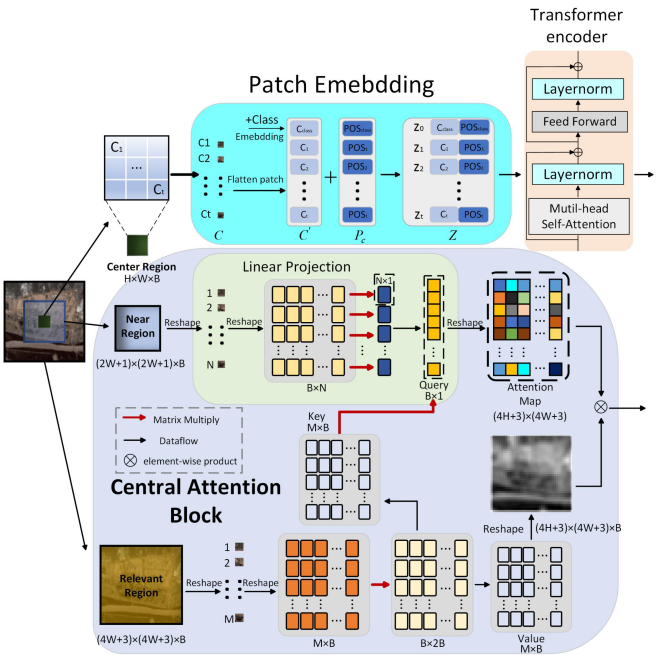

Fig. 4. Structure of details executed in different regions.

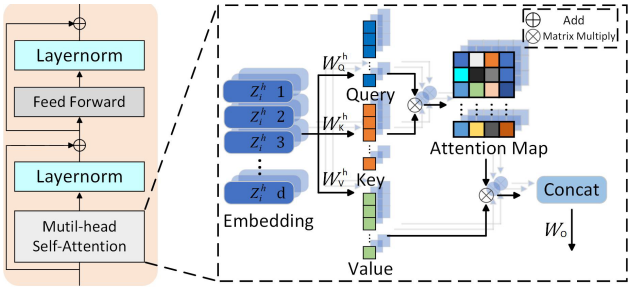

Fig. 5. Structure of multihead attention mechanism.

vectors, spatial location information is embedded into the network, resulting in the final input sequence $Z = P_c + C' = \left[\text{pos}_{\text{class}} + c_{\text{class}}, \text{pos}_1 + c_1, \ldots, \text{pos}_t + c_t\right] \in R^{U \times B}$, which is then input into the Transformer encoder for feature extraction.

Unlike traditional attention mechanisms, which model the correspondence between source and target sequences, the self-attention mechanism in Transformer establishes full connections within the sequence, enabling each position to interact with all other positions, effectively capturing long-range dependencies. As shown in Fig. 5, the input sequence to the multihead self-attention (MHSA) module is divided along the feature dimension into multiple subspaces. Each subspace is transformed through different weight matrices, allowing each attention head to learn diverse attention representations in different feature subspaces.

Let the input sequence be Z_i, with Z_i^h representing the input for one attention head. This input is mapped into query, key, and value vectors using three learnable projection matrices $\left[W_Q^h, W_K^h, W_V^h\right] \in R^{b \times b}$. The query vector is then dot-multiplied with the key vectors from other positions to measure the relevance between different pixels. The resulting correlation map is weighted and summed with the corresponding value vectors, aggregating the features within that subspace. Finally, the outputs of all attention heads are concatenated along the feature dimension and further integrated using a learnable linear transformation $W_O \in R^{B \times B}$. To enhance training stability, MHSA introduces residual connections and layer normalization between the input and output, as mathematically expressed

$$Q = Z_i^h W_Q^h, \quad K = Z_i^h W_K^h, \quad V = Z_i^h W_V^h \tag{3}$$

$$\text{Attention}^h(Z_i^h) = \text{soft max}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{4}$$

$$Z_i' = \text{LN}\left(\left[\text{Attention}^1; \ldots; \text{Attention}^H\right] W_O\right) + Z_i. \tag{5}$$

Here, Q, K, and V represent the query, key, and value vectors, respectively, and b is the input dimension of the hth attention head. softmax $(\cdot)$ denotes the Softmax activation function, $[\cdot]$ represents concatenation, H is the number of attention heads, and LN $(\cdot)$ refers to layer normalization. Subsequently, Z' is passed into a feed-forward network (FFN), consisting of two fully connected layers with GELU activation, enhancing the model's nonlinear expressive power. Similar to the previous module, FFN also includes layer normalization and residual connections, as expressed mathematically

$$Z_{i+1} = \text{LN}\left(\text{FFN}\left(Z_i'\right)\right) + Z_i'. \tag{6}$$

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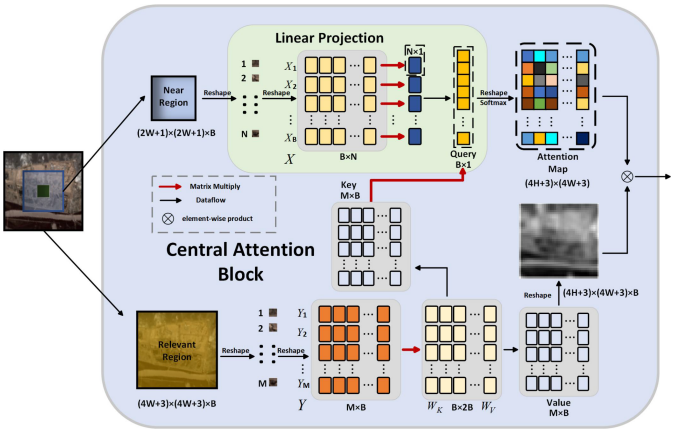

Fig. 6. Details structure of the CAB module.

These two steps form a complete Transformer encoder layer, which is repeated multiple times to progressively enhance the feature representation of the center region.

B. Central Attention Block

The local region around the center pixel often exhibits high spatial and spectral consistency. However, when class boundaries are ambiguous or mixed pixels are present, relying on single-scale information may hinder accurate classification. To address this, we further divide the patch into the near region and the relevant region. The near region, which surrounds the center region, consists of adjacent pixels with strong spatial continuity and category consistency. It provides local context features, helping to capture boundary changes and texture continuity, with an initial spatial size of $(2W+1)\times(2W+1)$. In contrast, the relevant region, with an initial size of $(4W+3)\times(4W+3)$, covers a broader area and contains rich global semantic information but is more prone to background noise or category mixing. Directly merging these two regions can lead to redundant or misleading features. Therefore, we propose the CAB, where the local discriminative information from the near region is used as a query to guide the model in selecting more discriminative global support features from the relevant region, enabling selective cross-scale feature extraction.

As shown in Fig. 6, to achieve cross-scale information alignment at the semantic level, we first perform semantic compression on the near region. Specifically, all pixels in the near region are flattened along the spatial dimension and transformed into a matrix $X = [X_1, X_2, \ldots, X_B] \in R^{B\times N}$, where $N = (2W+1)\times(2W+1)$. Subsequently, to enhance the positional semantic discrimination in cross-scale feature modeling, we introduce a refined linear mapping approach. For each position feature in the near region, an independent learnable parameter vector is used to map it to a 1-D query representation. The mapping process can be expressed as

$$Q_i = X_i W_i^Q, \quad i = 1, 2, \ldots, B \tag{7}$$

$$Q_{\mathrm{NR}} = [Q_1, Q_2, \ldots, Q_B]. \tag{8}$$

Here, $W_i^Q \in R^{N\times 1}$ represents the learnable projection vector for the ith position, and $Q_{\mathrm{NR}} \in R^{B\times 1}$ is the query representation derived from the near region. This design provides more flexible representation capabilities for different spatial positions. Although it introduces $B\times N$ additional parameters compared to using a unified $N\times 1$ projection, the spatial range of the near region is limited, so the computational and parameter overhead remains within acceptable bounds.

Subsequently, we process the relevant region following the self-attention paradigm. First, all pixels in the relevant region are flattened along the spatial dimension and reconstructed as $Y = [Y_1, Y_2, \ldots, Y_M] \in R^{M\times B}$, where $M = (4W+3)\times(4W+3)$. Two sets of learnable linear projection matrices, $W_K \in R^{B\times B}$ and $W_V \in R^{B\times B}$, are introduced to map the features to key and value, respectively, as expressed below

$$K_{RR} = YW_K, \quad V_{RR} = YW_V. \tag{9}$$

Here, K_{RR} and V_{RR} represent the key and value matrices, respectively. Next, the query vector Q_{NR} is dot-multiplied with all positions in the key matrix to obtain the cross-scale attention weight sequence. This sequence is then reshaped to match the input size of the relevant region, resulting in the attention map. The process can be expressed as

$$\mathrm{AM} = r(\mathrm{Soft\,max}(K_{RR}Q_{\mathrm{NR}})). \tag{10}$$

Here, r represents the reshape operation, and AM is the attention map. To apply the attention weights to the features, we reshape the value matrix to match the input size of the relevant region. Finally, the attention map is element-wise multiplied by the value matrix. The formula is as follows:

$$F = r(V_{RR}) \circ \mathrm{AM}. \tag{11}$$

Here, F represents the output of this module, and $\circ$ denotes element-wise multiplication. The above operation constitutes the complete CAB.

As shown in Fig. 3, after the features of the center region are extracted by multiple layers of Transformer encoders, the output vector c_{class} is used as the discriminative feature for the center region. For the near region and relevant region, after each output of the CAB, a convolution operation is applied to further integrate features and compress the spatial dimensions. After multiple iterations of this operation, global pooling is performed on the spatial scale, and the spectral dimension is used for discriminative features. Finally, the two 1-D features are concatenated and passed into the MLP for classification.

C. Comparative Learning Framework

The above process describes the complete training flow of the MSIA model under supervised conditions. To alleviate the issue of limited labeled samples, this article introduces a self-supervised contrastive learning strategy into the network training. By leveraging unlabeled hyperspectral samples to enhance the model's ability to capture and model latent spatial–spectral feature representations.

As shown in Fig. 7, this article adopts the SimSiam self-supervised contrastive learning framework without negative samples. The overall process includes two stages: pretraining and fine-tuning. In the pretraining stage, the original HSIs undergo data augmentation, using random rotations, occlusions, and cropping to generate two augmented views, x_i and

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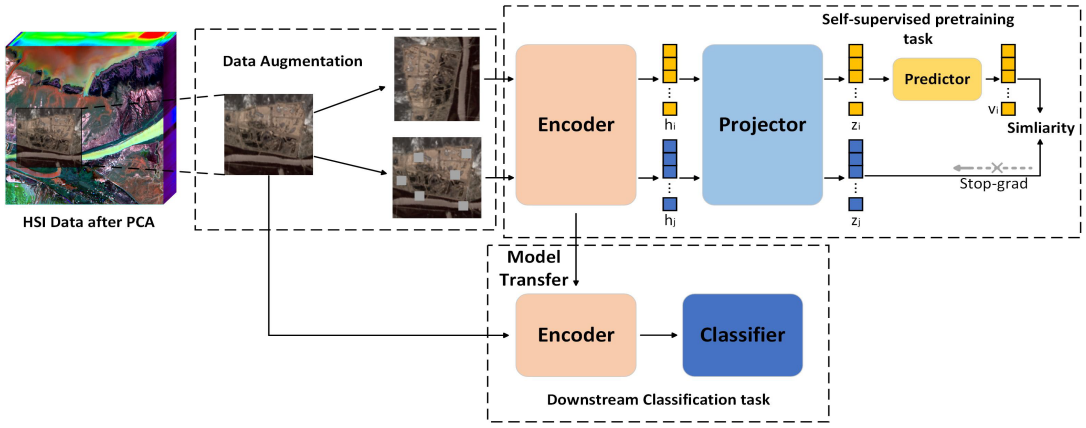

Fig. 7. Framework of the proposed method with contrastive learning method.

x_j, with different perspectives. These two images are then fed into the same encoder and projection head. The encoder, which removes the classifier from the MSIA architecture, is used to extract feature representations h_i and h_j. The projection head, consisting of two fully connected layers, maps the features to another space z_i and z_j, preventing the encoder from focusing too much on local details in the contrastive task, thus enabling it to freely learn general structural semantics. Subsequently, A two-layer fully connected predictor is added after one branch, while the gradient is stopped for the other branch. To enhance the model's robustness, both views are used for bidirectional matching as predictions and targets. The mathematical expression is as follows:

$$z_i = g(f(h_i)), \quad z_j = g(f(h_j)) \tag{12}$$

$$v_i = p(z_i), \quad v_j = p(z_j). \tag{13}$$

Here $f(\cdot)$ represents the encoder. $g(\cdot)$ the projection head, and $p(\cdot)$ the predictor (only on one branch). Finally, the loss function is constructed using negative cosine similarity, and its structure is as follows:

$$D(v, z) = -\frac{v^{\mathrm{T}} z}{\|v\| \cdot \|z\|} \tag{14}$$

$$L = \frac{1}{2}(D(v_i, \mathrm{sg}(z_j)) + D(v_j, \mathrm{sg}(z_i))). \tag{15}$$

Here $\mathrm{sg}(\cdot)$ represents stop-gradient. The loss function aims to maximize the cosine similarity between features from two perspectives. By blocking the gradient on one side, the model avoids converging to a trivial solution, enabling the contrastive objective to be achieved without explicitly constructing negative samples.

As shown in the lower part of Fig. 7, after completing self-supervised pretraining, the encoder parameters obtained from pretraining are transferred to the supervised training task. Then, a small number of labeled samples are used to fine-tune the entire MSIA model, which includes both the encoder and classifier, enabling the model to further adapt to the specific feature distribution of the classification task while preserving the general representations learned during pretraining.

TABLE I
LAND-COVER CATEGORY AND NUMBER OF TRAINING AND TEST SAMPLES ON IP DATASET

Class No.	Name	Train.	Test.
1	Alfalfa	2	44
2	Corn-notill	71	1357
3	Corn-mintill	42	788
4	Corn	12	225
5	Grass-pasture	24	459
6	Grass-trees	37	693
7	Grass-pasture-mowed	1	27
8	Hay-windrowed	24	454
9	Oats	1	19
10	Soybean-notill	49	923
11	Soybean-mintill	123	2332
12	Soybean-clean	30	563
13	Wheat	10	195
14	Woods	63	1202
15	Buildings-Grass-Trees-Drives	19	367
16	Stone-Steel-Towers	4	88
	Total	512	9737

IV. EXPERIMENTS

To validate the effectiveness of the proposed MSIA network and contrastive pretraining strategy, comprehensive experiments were conducted on five datasets. All experiments were performed under the same hardware configuration, including an AMD Ryzen 5 5600 six-core processor, 16 GB of RAM, NVIDIA GeForce RTX 3060 Ti GPU, PyTorch 1.12.0, and CUDA 11.3.

A. Data Description

This study selected three representative public datasets—Indian Pines (IPs), Pavia University (PU), Salinas (SAs), and Houston2018 (HU18)—along with a self-labeled Yellow River Estuary coastal wetland (YRE) dataset to evaluate the method's generalization under varying geographic and land cover complexities. Tables I–V summarize the land cover classes and the number of training and testing samples for each dataset.

1) The IPs dataset is one of the earliest benchmarks for HSI classification. It was collected by the Airborne

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TABLE II
LAND-COVER CATEGORY AND NUMBER OF TRAINING AND TEST SAMPLES ON PU DATASET

Class No.	Name	Train.	Test.
1	Asphalt	66	6565
2	Meadows	186	18463
3	Gravel	21	2078
4	Trees	30	3034
5	Painted metal sheets	13	1332
6	Bare Soil	50	4979
7	Bitumen	13	1317
8	Self-Blocking Bricks	36	3646
9	Shadows	9	938
	Total	424	42352

TABLE III
LAND-COVER CATEGORY AND NUMBER OF TRAINING AND TEST SAMPLES ON THE SA DATASET

Class No.	Name	Train.	Test.
1	Brocoli green weeds1	20	1989
2	Brocoli green weeds 2	37	3689
3	Fallow	19	1957
4	Fallow rough plow	14	1380
5	Fallow smooth	26	2652
6	Stubble	39	3920
7	Celery	35	3544
8	Grapes untrained	112	11159
9	Soil vinyard develop	62	6141
10	Com senesced green weeds	32	3246
11	Lettuce romaine 4 wk	10	1058
12	Lettuce romaine 5 wk	19	1908
13	Lettuce romaine 6 wk	9	907
14	Lettuce romaine 7 wk	10	1060
15	Vinyard untrained	72	7196
16	Vinyard vertical trellis	18	1789
	Total	534	53595

TABLE IV
LAND-COVER CATEGORY AND NUMBER OF TRAINING AND TEST SAMPLES ON THE YRE DATASET

Class No.	Name	Train.	Test.
1	Mudflat	7	3576
2	Bare soil	11	5375
3	Tamarix	7	3575
4	Suaeda salsa	8	4235
5	Spartina alterniflora	47	23488
6	Turbid water	34	16987
7	Tidal flat reed	12	5975
8	Clear water	24	12229
9	Bare lake beach	8	4137
	Total	158	79574

TABLE V
LAND-COVER CATEGORY AND NUMBER OF TRAINING AND TEST SAMPLES ON HU18 DATASET

Class No.	Name	Train.	Test.
1	Healthy grass	29	9770
2	Stressed grass	98	32404
3	Synthetic grass	2	682
4	Evergreen grass	41	13554
5	Deciduous grass	15	5006
6	Soil	14	4502
7	Water	2	264
8	Residential	119	39653
9	Commercial	671	223081
10	Road	138	45728
11	Sidewalk	102	33927
12	Crosswalk	5	1513
13	Major Thoroughfares	139	46209
14	Highway	30	9835
15	Railway	21	6916
16	Paved Parking Lot	35	11465
17	Gravel Parking Lot	2	144
18	Cars	20	6527
19	Trains	16	5353
20	Seats	20	6804
	Total	1519	503337

Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 over agricultural fields in Indiana, USA, covering a 145 × 145 pixel region with a spatial resolution of 20 m. AVIRIS records data across 220 spectral bands from 0.4 to 2.5 μm, of which 200 bands are retained after removing those affected by water absorption (bands 104–108, 150–163, and 220). The dataset includes 10 249 labeled pixels across 16 crop-related classes. Its highly imbalanced class distribution and similar spectral signatures pose significant classification challenges. We used 5% of the data for training, with the remainder reserved for testing (see Table I).

2) The PU dataset is an HSI captured in 2003 over Pavia, Italy, using German Reflective Optics System Imaging Spectrometer. It covers 115 spectral bands in the 0.43–0.86 μm range with a spatial resolution of 1.3 m. Due to noise contamination, 12 bands are discarded, resulting in 103 usable bands. The image size is 610 × 340, with 42 776 labeled pixels across 9 land cover classes. We used 1% of the data for training, with the remainder reserved for testing (see Table II).

3) The SAs dataset was collected by the AVIRIS sensor over agricultural fields in California's SAs Valley. It covers the 0.4–2.5 μm spectral range with a spatial resolution of 3.7 m. The image size is 512 × 217, containing 224 spectral bands, of which 204 are retained after removing noisy ones. After excluding background pixels, 54 129 labeled samples remain, spanning 16 land cover classes. We used 1% of the data for training, with the remainder reserved for testing (see Table III).

4) The Yellow River Estuary coastal wetland dataset was acquired in 2019 by the hyperspectral sensor onboard China's Gaofen-5 (GF-5) satellite over Dongying, Shandong Province, covering the core area of the Yellow River Delta National Nature Reserve. The image includes 330 spectral bands spanning 390–1029 nm (VNIR) and 1005–2513 nm (SWIR), with 50 noisy bands removed, resulting in 280 bands used for classification. Originally at 30 m spatial resolution, the

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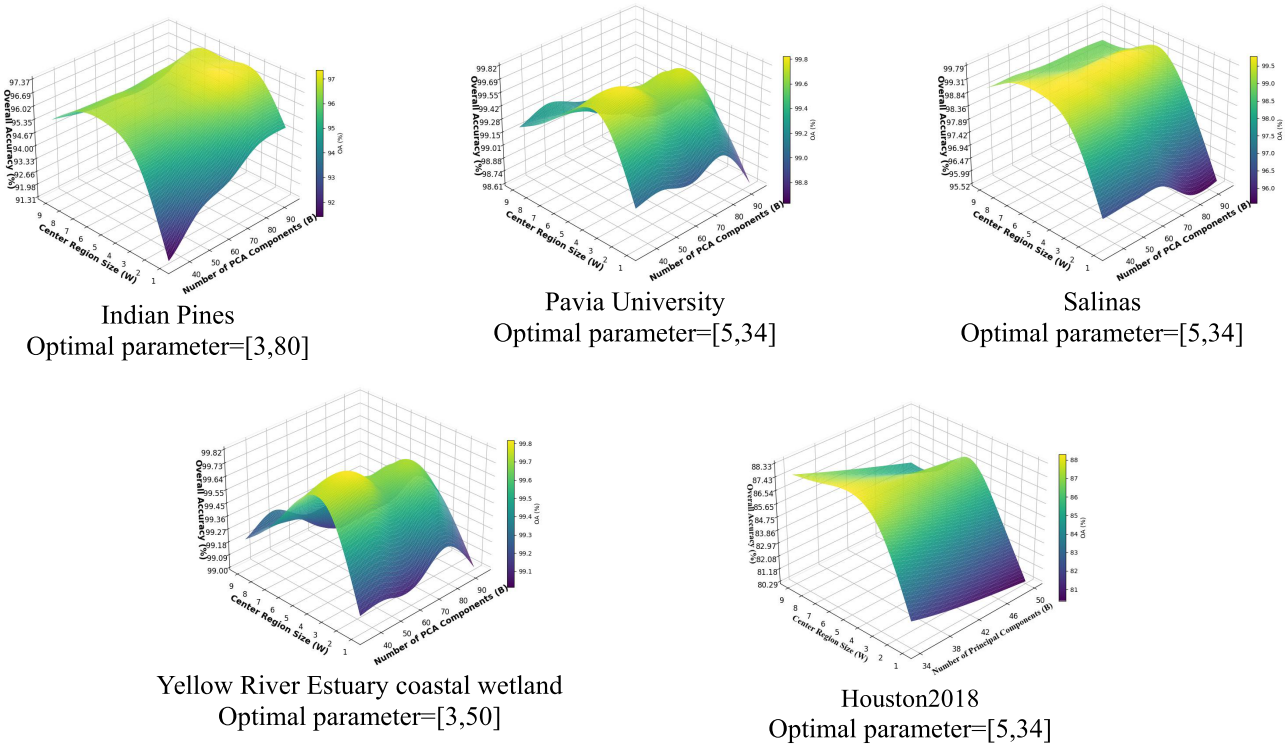

Fig. 8. Impact of center region size and number of principal components on classification accuracy.

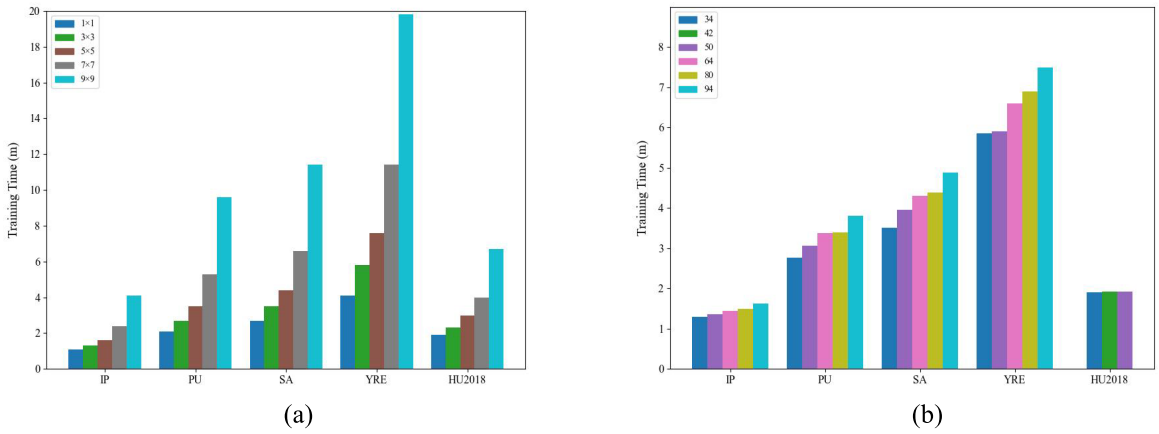

Fig. 9. (a) Impact of center region size on training time. (b) Impact of the number of principal components on training time.

data were pansharpened using Landsat 8 panchromatic imagery to enhance spatial detail to 15 m. The image size is 710 × 673 pixels, with 79 732 labeled samples across 9 land cover classes. We used 0.2% of the data for training, with the remainder reserved for testing (see Table IV).

5) The Houston 2018 dataset provided by the IEEE GRSS DASE initiative was collected over the University of Houston campus and surrounding areas. It is a multisensor dataset containing hyperspectral data (1)-m resolution) from 48 bands and LiDAR data from three bands. The dataset contains 504 856 labeled pixels categorized into 20 typical land-cover classes. We used 0.3% of the data for training, with the remainder reserved for testing (see Table V).

B. Comparison Methods

To verify the effectiveness of the proposed method and contrastive pretraining strategy, several representative state-of-the-art HSI classification models were selected for comparison, including 3D-CNN [19], SSRN [21], HybridSN [22], VIT [26], SpectralFormer [27], SSFTT [33], and MSIA-WP.

1) *3D-CNN* performs joint spectral–spatial modeling using a classical convolutional structure with a kernel size of $7 \times 3 \times 3$.
2) *SSRN* adopts a dual-branch residual network to separately extract spectral and spatial features using kernels of sizes $7 \times 1 \times 1$ and $1 \times 3 \times 3$, respectively.
3) *HybridSN* integrates 3-D and 2-D convolutions, with three 3-D conv layers, one 2-D conv layer, three fully connected layers, and two dropout layers.

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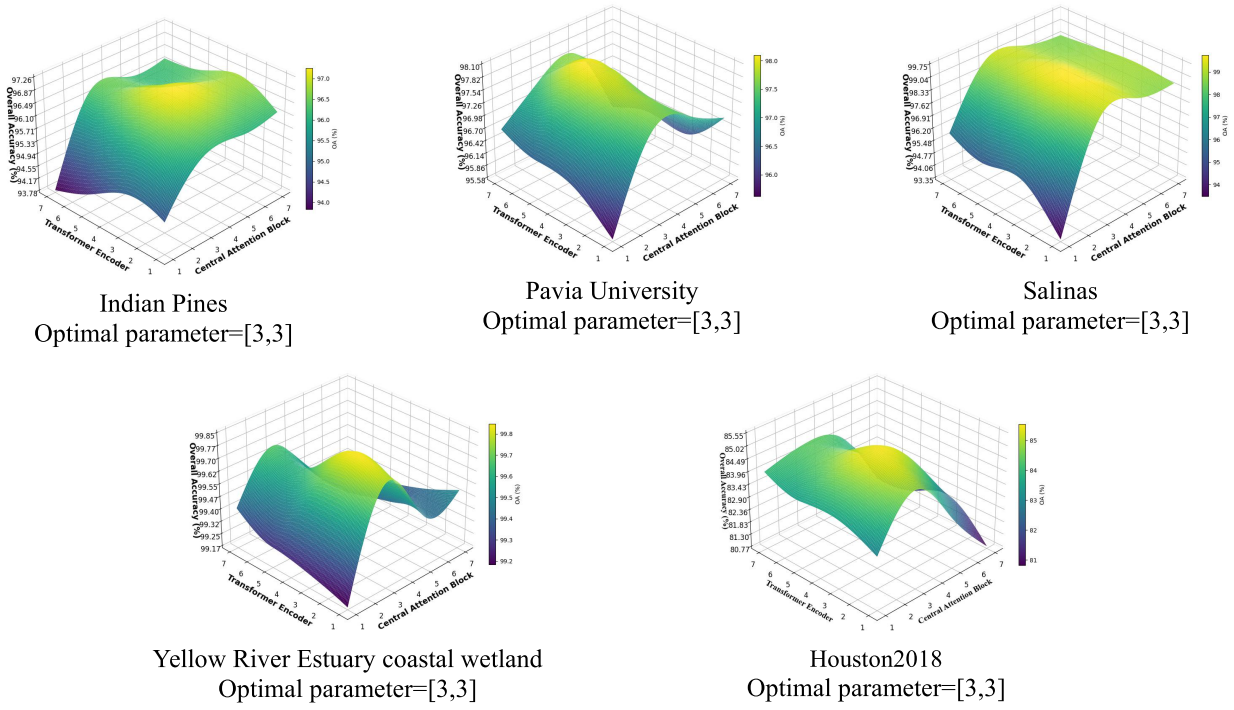

Fig. 10. Impact of depth of transformer encoder and CAB on classification accuracy.

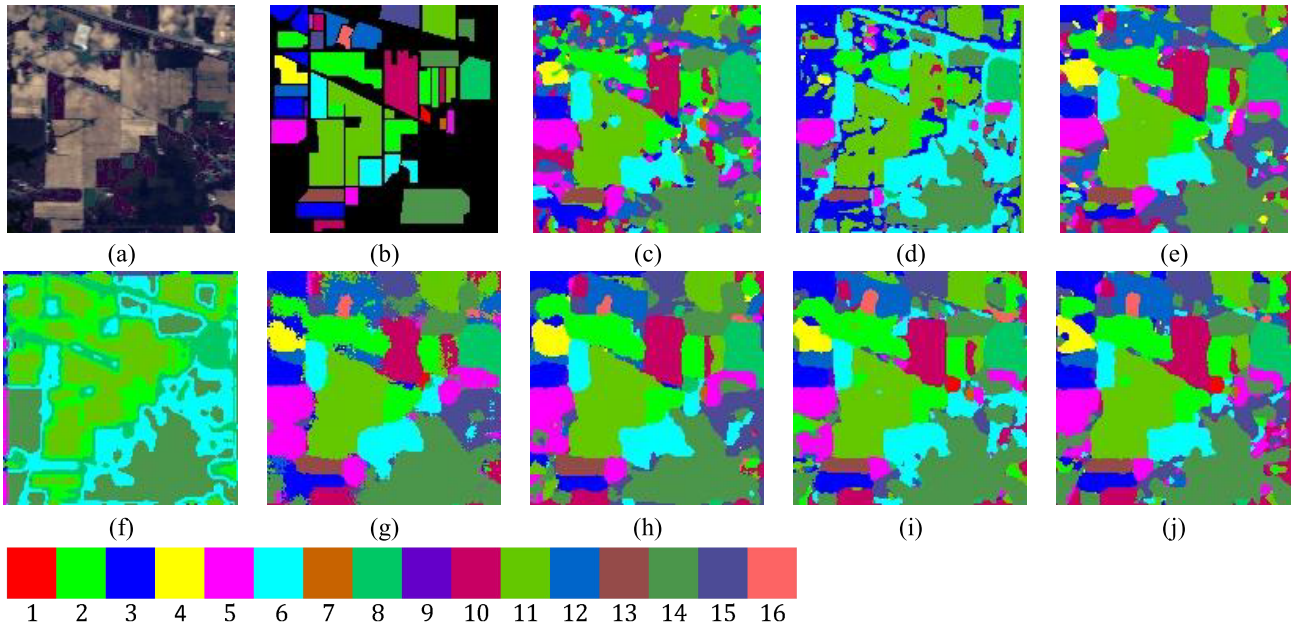

Fig. 11. Classification maps of the IPs dataset. (a) False-color image. (b) Ground-truth map. (c) 3D-CNN. (d) SSRN. (e) HybridSN. (f) VIT. (g) SpectralFormer. (h) SSFTT. (i) MSIA-WP. (j) MSIA.

4) *VIT* applies the original vision Transformer architecture to HSI classification, using six transformer encoder layers for feature extraction.
5) *SpectralFormer* is a pure transformer-based method for HSI classification. It captures local spectral sequences via intragroup spectral embedding and employs skip connections to mitigate deep-layer information loss.
6) *SSFTT* combines 3-D and 2-D convolutions to extract local features before passing them to a transformer encoder for long-range modeling.
7) *MSIA-WP* denotes the proposed MSIA model without pretraining, used to evaluate the benefit of the contrastive pretraining strategy.

To ensure a fair comparison, all models were trained using the Adam optimizer with a batch size of 128 and a learning rate of 0.0001. The number of training epochs was set to 100 for the IP dataset and 60 for the others. All remaining hyperparameters followed the best settings reported in the original papers. For MSIA, the pretraining stage used 1000 epochs.

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TABLE VI
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR THE IP DATASET

Class	3D-CNN	SSRN	HybridSN	VIT	SpectralFormer	SSFTT	MSIA-WP	MSIA
1	60.00±54.77	0.00 ±0.00	79.52±44.47	0.00±0.00	97.50±3.73	96.96 ±2.85	98.78 ±2.74	**100.00 ±0.00**
2	85.30±4.25	75.07±27.97	83.99±8.50	34.11±1.35	91.71±1.68	94.02±1.57	92.01±2.86	**96.19±2.94**
3	79.50±3.74	65.10±18.59	92.97±2.28	60.25±7.86	89.64±2.17	92.22±2.25	94.09±3.12	**95.20±1.79**
4	90.20±4.32	36.90±35.46	90.42±6.46	30.76±4.97	99.41±0.54	**100.00±0.00**	99.40±1.10	98.29±1.00
5	90.76±3.59	90.11±7.90	94.35±2.96	**100.00±0.00**	91.13±4.88	95.06±1.84	97.91±0.33	99.40±0.63
6	95.78±2.83	88.95±7.77	97.05±2.19	51.58±1.65	98.28±0.74	97.90±1.63	99.10±0.33	**99.62±0.60**
7	99.00±2.24	00.00±00.00	60.00±54.77	00.00±00.00	**100.00±00.00**	**100.00±0.00**	**100.00±00.00**	98.00±4.47
8	91.35±2.65	91.66±6.20	91.38±2.50	54.78±8.80	96.09±2.24	96.45±3.06	98.58±1.72	**99.25±0.43**
9	98.75±2.80	0.00±0.00	60.00±54.77	0.00±0.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	100.00±0.00
10	84.21±5.56	75.24±14.46	94.82±2.87	0.00±0.00	86.63±1.97	92.85±5.66	**94.42±2.29**	94.25±1.99
11	89.56±2.04	62.62±4.87	93.63±1.68	45.71±2.39	91.92±2.93	94.02±1.16	**96.58±1.08**	96.43±1.54
12	73.66±7.85	61.79±38.65	86.00±7.33	28.89±44.17	92.33±2.88	95.06±2.42	**96.92±1.30**	96.12±2.11
13	95.74±3.68	94.63±4.38	96.82±5.48	60.00±54.77	98.35±0.26	98.66±1.04	99.90±0.22	**100.00±0.00**
14	93.87±1.69	93.57±4.35	95.45±1.32	77.17±2.05	98.01±1.18	**98.86±0.83**	98.60±0.66	98.50±1.36
15	81.24±5.75	90.26±7.93	90.51±5.24	93.60±11.75	97.04±0.48	95.54±1.19	98.14±1.85	**98.81±0.77**
16	86.40±8.29	**100.00±0.00**	95.19±4.66	0.00±0.00	93.52±2.86	94.15±6.04	88.11±3.66	91.47±7.11
OA (%)	87.51±1.38	68.25±5.73	91.89±1.26	49.91±0.54	93.04±1.68	95.13±0.87	96.19±0.35	**96.96±0.40**
AA (%)	77.86±3.66	52.15±5.26	76.51±5.12	30.48±0.68	82.43±2.92	89.42±2.71	93.36±1.51	**93.80±1.16**
$k \times 100$	85.73±1.58	63.10±6.56	90.72±1.44	40.74±0.86	92.06±1.94	94.43±1.00	95.65±0.39	**96.53±0.46**

TABLE VII
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR PU DATASET

Class	3D-CNN	SSRN	HybridSN	VIT	SpectralFormer	SSFTT	MSIA-WP	MSIA
1	88.02±5.52	82.09±2.27	91.30±1.84	64.93±1.65	92.64±0.76	95.09±0.79	97.33±1.14	**97.47±0.75**
2	96.58±0.72	92.92±3.69	97.91±0.41	81.05±1.52	98.14±0.17	98.91±0.40	98.93±0.41	**99.29±0.13**
3	88.79±3.83	74.73±26.14	91.37±5.27	0.00±0.00	92.97±2.77	96.68±2.88	98.70±0.83	**99.02±0.18**
4	97.52±1.43	97.96±4.49	98.62±0.83	97.42±2.00	98.34±0.59	97.02±0.32	98.35±0.73	**99.03±0.23**
5	99.47±0.51	99.34±1.13	98.84±0.38	96.25±2.95	99.38±0.51	99.56±0.09	99.48±0.40	**99.57±0.19**
6	95.63±3.36	92.20±5.30	96.74±1.50	57.19±5.74	97.93±0.54	97.60±1.37	99.44±0.57	**99.53±0.34**
7	92.41±3.65	41.59±45.41	90.63±3.89	35.12±32.69	96.23±4.16	97.38±1.35	**99.05±0.55**	98.83±0.93
8	77.82±3.88	74.36±13.29	79.85±6.32	47.09±3.42	85.56±1.88	86.60±2.30	87.45±2.64	**89.87±0.94**
9	91.24±5.01	90.78±4.95	95.13±5.45	**96.54±0.81**	96.47±0.39	96.34±1.37	96.40±0.79	94.85±0.30
OA (%)	92.92±1.68	88.40±1.93	94.50±0.92	74.18±1.62	95.86±0.29	96.75±0.26	97.55±0.60	**98.03±0.14**
AA (%)	89.76±2.35	77.05±3.11	90.56±1.23	59.98±2.15	92.93±0.46	94.98±0.32	95.49±1.06	**96.31±0.26**
$k \times 100$	90.57±2.24	84.36±2.66	92.68±1.22	64.25±2.29	94.48±0.38	95.68±0.34	96.75±0.79	**97.38±0.18**

TABLE VIII
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR THE SA DATASET

Class	3D-CNN	SSRN	HybridSN	VIT	SpectralFormer	SSFTT	MSIA-WP	MSIA
1	99.98±0.04	89.83±18.01	**100.00±0.00**	38.18±52.38	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	99.34±0.36	91.06±13.80	**99.98±0.02**	64.12±5.73	99.64±0.21	99.97±0.06	99.95±0.03	99.98±0.04
3	99.39±0.59	53.12±40.20	98.98±1.02	25.83±38.67	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
4	99.38±0.44	92.68±5.72	**99.41±0.53**	78.74±44.05	95.16±1.36	98.26±0.57	98.87±0.20	98.69±1.21
5	97.39±0.96	85.93±13.42	99.41±0.24	56.49±2.67	98.77±0.12	**99.50±0.14**	98.85±0.24	99.37±0.29
6	98.55±1.09	99.91±0.16	99.80±0.42	93.40±1.94	99.81±0.29	**99.97±0.06**	99.97±0.07	99.95±0.11
7	99.59±0.62	97.66±1.93	**100.00±0.00**	90.66±10.13	99.99±0.03	**100.00±0.00**	99.99±0.01	**100.00±0.00**
8	95.34±1.26	80.42±7.17	97.08±0.51	71.60±2.82	98.12±0.40	98.52±0.44	99.93±0.05	**99.94±0.08**
9	99.80±0.09	91.41±8.57	99.76±0.27	72.65±5.52	99.95±0.04	**100.00±0.00**	**100.00±0.00**	99.99±0.01
10	99.78±0.15	74.96±16.10	99.61±0.23	82.78±12.75	99.52±0.11	99.30±0.74	**99.93±0.06**	99.78±0.18
11	95.59±0.67	38.73±53.03	97.03±1.49	10.00±22.36	97.53±1.41	98.58±0.62	**99.42±0.54**	98.68±0.82
12	99.31±0.61	50.26±31.04	99.41±0.50	76.82±5.26	98.74±0.96	99.70±0.33	99.68±0.36	**99.70±0.19**
13	95.04±4.29	47.93±48.88	98.84±1.05	95.64±4.18	99.86±0.30	**99.96±0.06**	99.89±0.13	99.87±0.29
14	97.69±0.66	59.14±53.99	99.18±0.40	98.33±1.48	97.32±1.63	98.41±0.60	98.84±0.44	**99.50±0.60**
15	91.91±1.23	79.43±5.92	93.71±0.76	60.88±2.68	95.89±1.62	97.57±0.12	99.45±0.25	**99.48±0.23**
16	99.73±0.37	99.06±1.19	99.98±0.05	95.22±1.60	99.97±0.08	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	97.28±0.48	83.44±3.47	98.27±0.18	73.12±0.48	98.63±0.29	99.18±0.05	99.77±0.03	**99.79±0.02**
AA (%)	97.94±0.64	75.23±5.16	99.01±0.21	63.03±1.53	98.93±0.12	99.44±0.07	99.67±0.03	**99.71±0.04**
$k \times 100$	96.98±0.54	81.46±3.93	98.07±0.21	69.76±0.57	98.48±0.32	99.08±0.06	99.74±0.04	**99.76±0.03**

C. Parameter Analysis

Hyperparameters play a critical role in determining model performance. To achieve optimal results, key hyperparameters were systematically tuned using supervised experiments with training and testing samples consistent with Tables I–IV.

1) *Center Region Size* (W) *and Number of Principal Components* (B): The patch size and the number of retained

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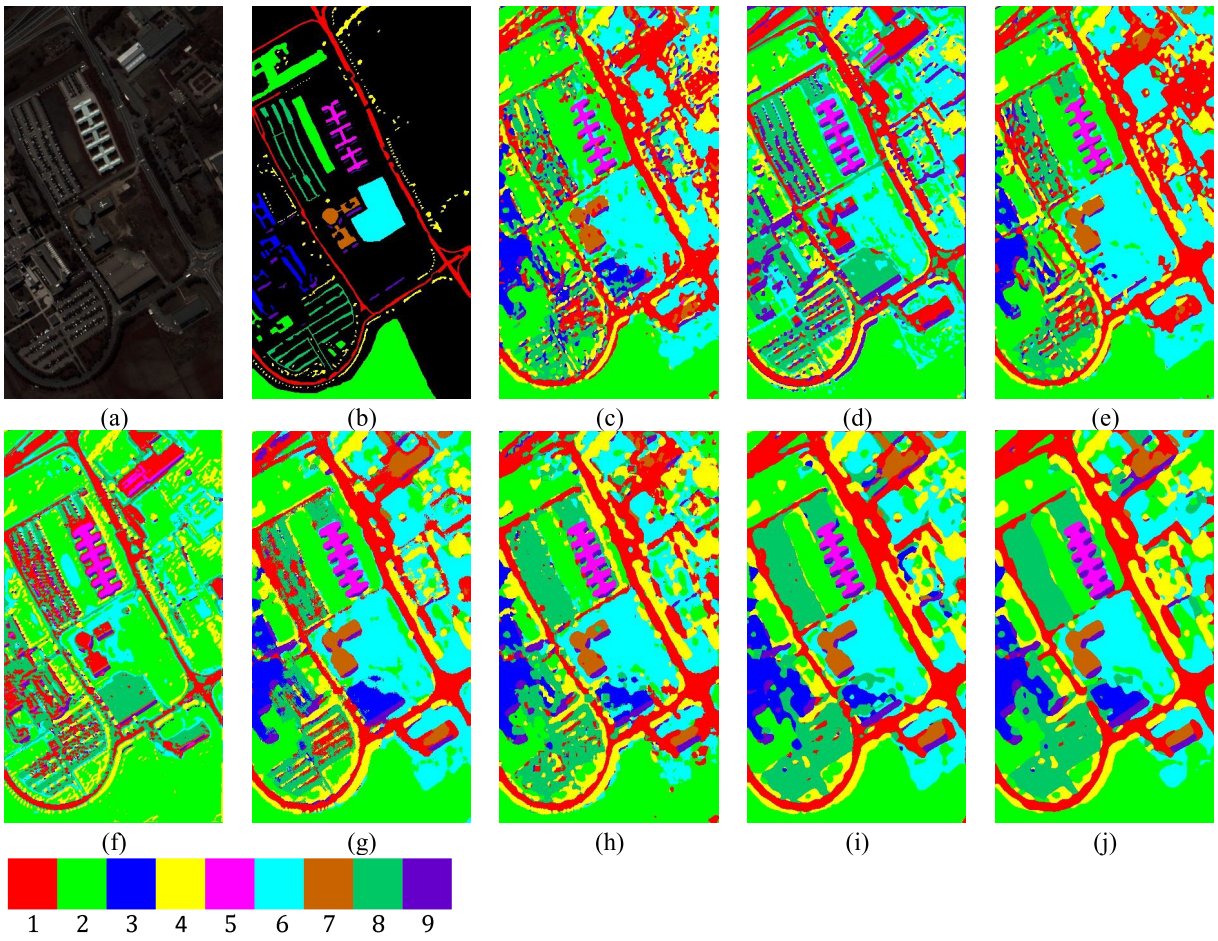

Fig. 12. Classification maps of the PU dataset. (a) False-color image. (b) Ground-truth map. (c) 3D-CNN. (d) SSRN. (e) HybridSN. (f) VIT. (g) SpectralFormer. (h) SSFTT. (i) MSIA-WP. (j) MSIA.

TABLE IX
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR THE YRE DATASET

Class	3D-CNN	SSRN	HybridSN	VIT	SpectralFormer	SSFTT	MSIA-WP	MSIA
1	93.77±2.04	97.28±3.50	98.15±0.86	95.04±4.13	**99.64±0.81**	99.63±0.33	97.53±0.88	97.06±2.19
2	88.91±6.21	99.96±0.03	97.49±2.06	77.52±10.38	97.97±2.61	97.76±3.23	**100.00±0.00**	99.94±0.05
3	87.05±9.03	95.82±3.13	94.40±3.17	98.14±0.41	98.29±1.76	98.14±1.51	97.81±4.65	**100.00±0.00**
4	96.54±1.35	96.70±6.20	92.86±3.78	89.52±1.24	91.42±4.80	96.12±2.89	**98.70±1.73**	96.70±1.11
5	97.78±1.06	96.79±1.02	98.86±0.85	97.69±0.65	98.33±1.20	99.50±0.21	99.59±0.25	**99.90±0.05**
6	95.66±2.23	99.90±0.07	97.30±0.75	83.38±0.47	99.49±0.32	99.16±0.27	**99.94±0.06**	99.89±0.05
7	82.59±6.39	96.58±4.68	97.76±2.26	**99.89±0.07**	99.45±0.12	99.69±0.25	99.41±0.64	99.58±0.49
8	93.11±2.57	99.89±0.10	99.16±0.55	97.44±3.97	99.51±0.54	99.81±0.06	99.33±0.61	**99.95±0.09**
9	88.14±9.25	99.36±0.71	93.43±3.56	98.46±0.30	98.48±1.96	99.96±0.07	**100.00±0.00**	99.99±0.01
OA (%)	93.62±1.55	98.19±0.75	97.51±0.43	91.66±0.64	98.45±0.36	99.14±0.30	99.43±0.23	**99.58±0.12**
AA (%)	89.92±2.62	96.06±1.79	96.73±0.90	81.60±1.27	97.17±0.72	98.70±0.68	99.02±0.44	**99.39±0.12**
$k \times 100$	92.24±1.89	97.79±0.92	96.98±0.52	89.76±0.78	98.12±0.43	98.95±0.36	99.30±0.28	**99.49±0.15**

principal components in PCA determine the input information volume and are critical to model performance. As the patch is centered on the center region, we focus on the influence of the center region size. Experiments on all five datasets varied the region size from 1×1 to 9×9 and the number of retained PCs from 34 to 94 (HU18 is 34–50). As shown in Fig. 8, both hyperparameters notably affect model performance. When the patch size is small, the number of PCA has a stronger influence on the model performance. In particular, on the IP dataset, which contains numerous classes with high spectral overlap, the accuracy difference caused by PCA reaches up to 6%. As the window size increases, the sensitivity to PCA dimensionality becomes less pronounced, and the variation in accuracy gradually stabilizes. Within an appropriate range (e.g., $W = 3$–7), the accuracy deviation remains within 0.5% even at extreme PCA values. The central region size exhibits a more dominant effect on performance. For the IP and HU18 datasets, the accuracy difference between the

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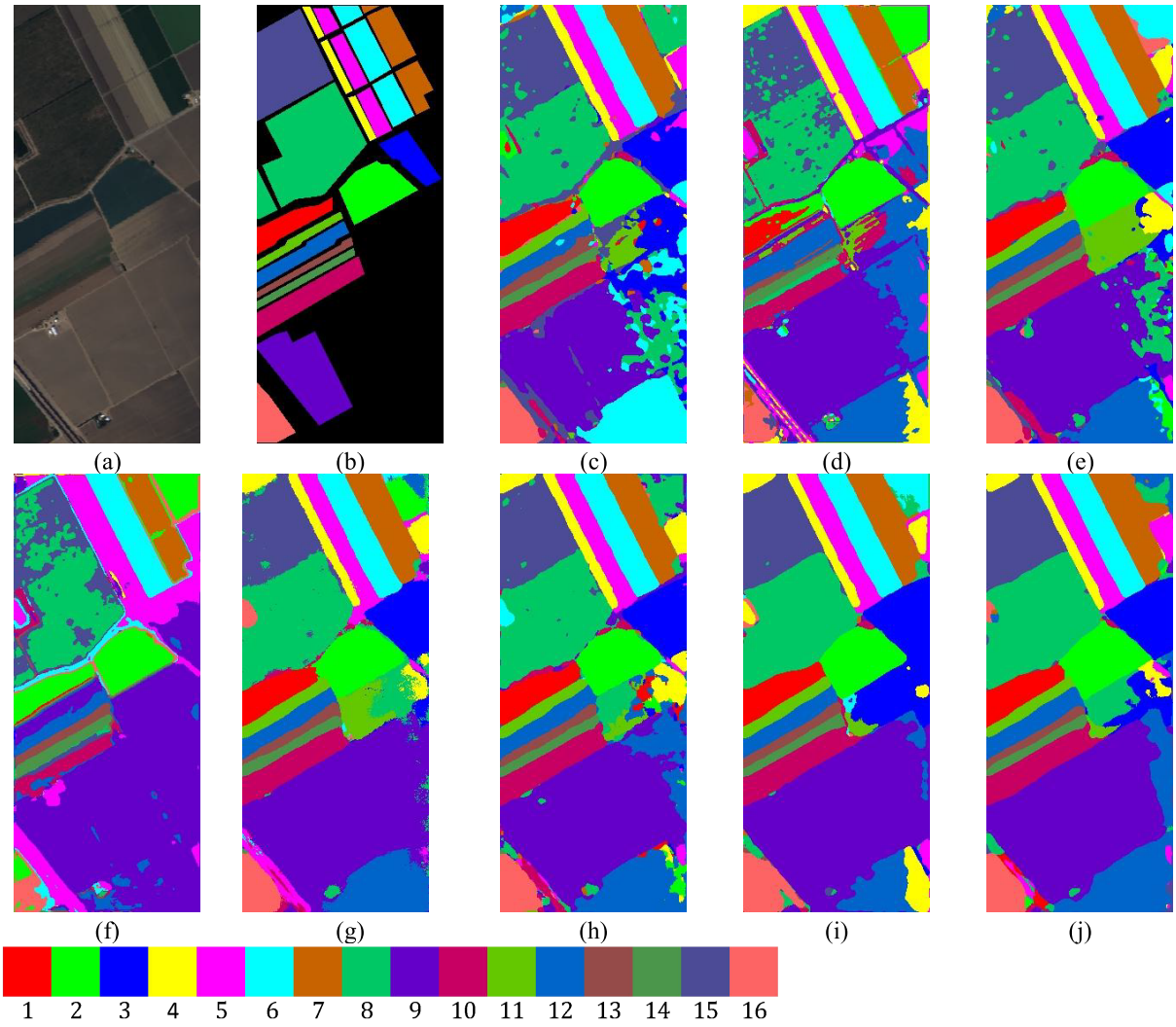

Fig. 13. Classification maps of the SAs dataset. (a) False-color image. (b) Ground-truth map. (c) 3D-CNN. (d) SSRN. (e) HybridSN. (f) VIT. (g) SpectralFormer. (h) SSFTT. (i) MSIA-WP. (j) MSIA.

optimal and extreme settings exceeds 5%. Moreover, the degradation caused by contextual information loss in a small patch is more severe than that caused by redundant information in a large patch. In the stable region (W = 3–7), the accuracy fluctuation is limited to less than 1%.

In addition, Fig. 9 records the impact of different parameters on training time. The results indicate that the center window size has a more significant effect, exhibiting a nonlinear increase in training time. In contrast, increasing the number of retained PCA components results in a more stable, approximately linear growth in computational cost.

2) *Depth of Transformer Encoder and CAB:* As the two core modules in the proposed architecture, the depth of the Transformer encoder and CAB directly affects both feature representation capability and computational cost. We conducted experiments on all five datasets, varying their depths from 1 to 7. As shown in Fig. 10, increasing the depth of the Transformer encoder from 1 to 3 significantly enhances classification performance across all datasets. For example, on the IPs dataset, OA improves from 94.69% at depth [1, 1] to 97.10% at [3, 3]. However, further increasing the depth to 5 or 7 leads to diminishing returns or slight degradation, likely due to feature redundancy or overfitting. A similar trend is observed for the CAB. When its depth increases from 1 to 3, most datasets benefit from enhanced classification accuracy, indicating that moderate cross-scale interactions improve local-global feature synergy. However, beyond three layers, additional stacking introduces redundant or irrelevant information, potentially weakening the influence of the center region. Overall, the optimal configuration across all datasets is achieved when both the Transformer encoder and CAB have a depth of 3.

D. Classification Results and Analysis

The classification maps obtained by all methods are shown in Figs. 11–15, and the corresponding quantitative evaluation results are summarized in Tables VI–X (with the best results highlighted in bold).

1) *IPs dataset:* Under the IPs dataset, with only 5% labeled samples, significant performance disparities emerged.

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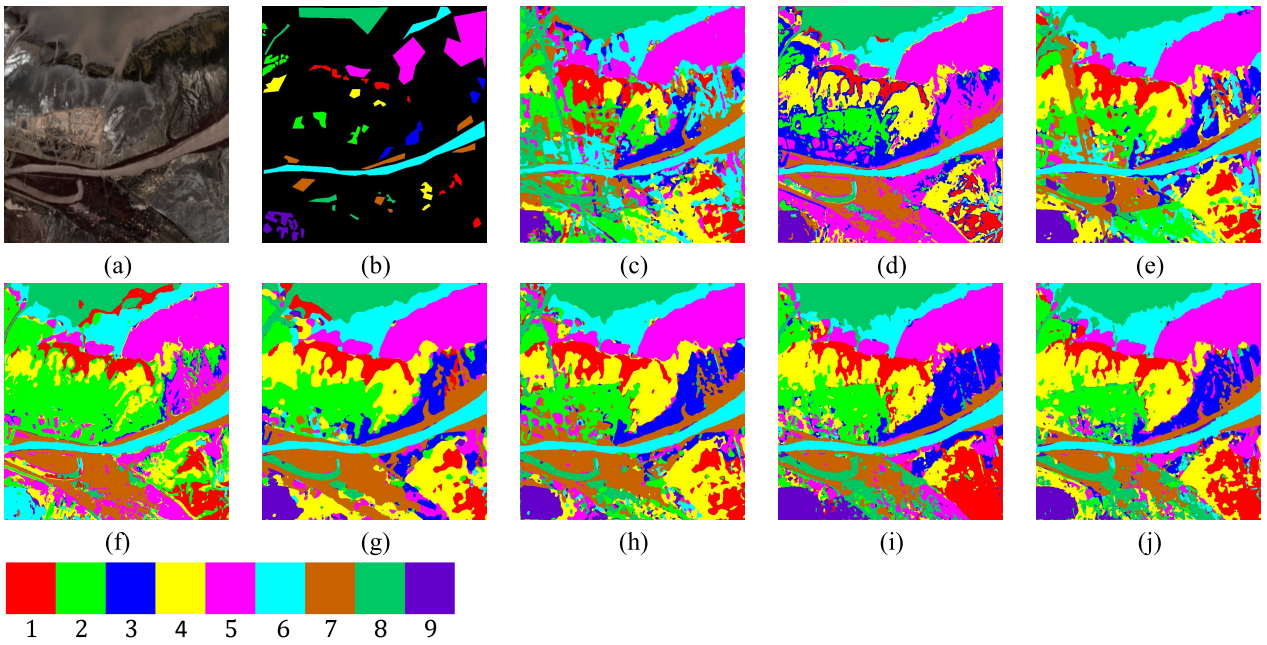

Fig. 14. Classification maps of the Yellow River Estuary coastal wetland dataset. (a) False-color image. (b) Ground-truth map. (c) 3D-CNN. (d) SSRN. (e) HybridSN. (f) VIT. (g) SpectralFormer. (h) SSFTT. (i) MSIA-WP. (j) MSIA.

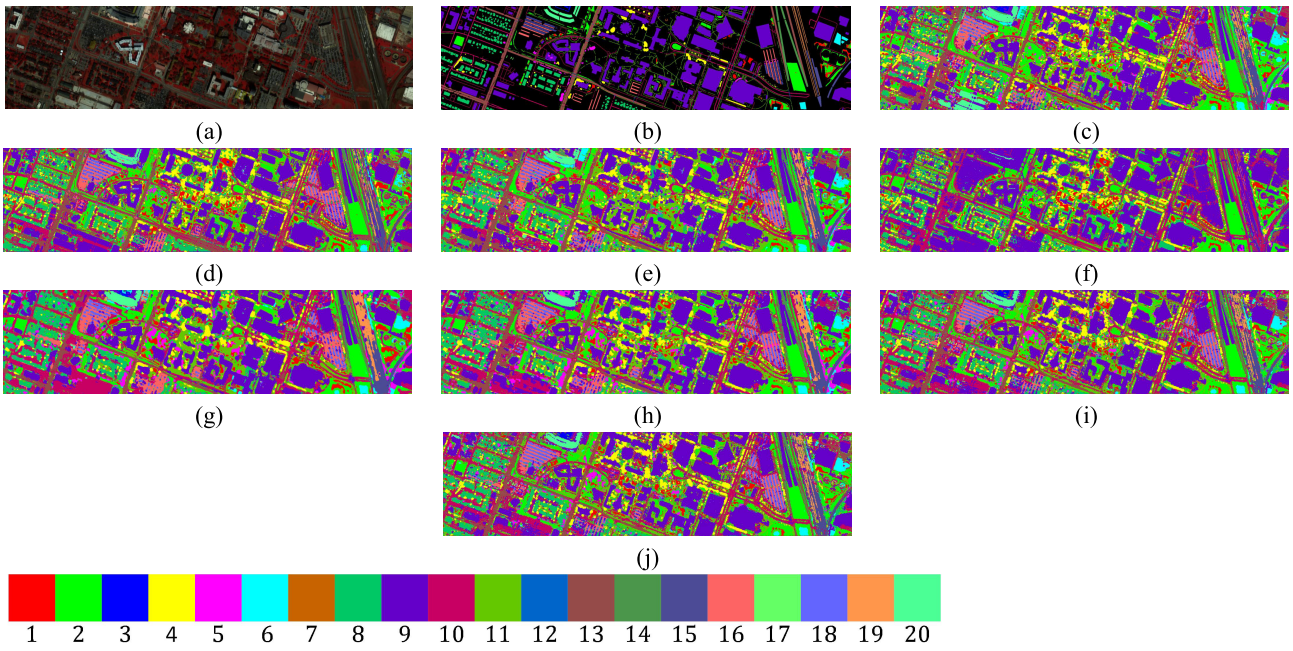

Fig. 15. Classification maps of the HU18 dataset. (a) False-color image. (b) Ground-truth map. (c) 3D-CNN. (d) SSRN. (e) HybridSN. (f) VIT. (g) SpectralFormer. (h) SSFTT. (i) MSIA-WP. (j) MSIA.

Convolution-based models severely overfitted: 3D-CNN and HybridSN showed accuracy for Alfalfa (Class 1) fluctuated near 50%, while SSRN failed completely on Alfalfa (Class 1), Grass-pasture-mowed (Class 7), and Oats (Class 9) across all five runs. This is mainly attributed to: 1) the large parameter size of CNNs relative to the small number of samples, which amplifies gradient noise; and 2) the local receptive field limitation of CNNs, which fails to capture spectral–spatial variations of the same class (e.g., the boundary versus center pixels of Corn-notill fields), leading to divergent parameter updates. Among Transformer-based methods, most performed significantly better than the vanilla ViT, whose OA dropped to 49.91%, with several categories completely misclassified. This is likely due to the flattening of 3-D cubes into 2-D sequences, which disrupts spatial–spectral correlations. In contrast, SpectralFormer achieved much higher accuracy owing to its tailored spectral embedding and attention design. SSFTT and MSIA-WP showed comparable overall performance, but MSIA-WP achieved a nearly 4% higher AA, indicating stronger generalization,

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TABLE X
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR HU18 DATASET

Class	3D-CNN	SSRN	HybridSN	VIT	SpectralFormer	SSFTT	MSIA-WP	MSIA
1	69.10±0.64	81.70±7.91	69.96±5.77	79.42±9.69	70.40±2.27	72.60±4.50	**82.30±5.46**	74.12±1.75
2	78.00±1.97	**90.80±2.62**	82.46±4.06	74.66±6.29	88.16±2.02	87.64±1.47	87.64±2.41	90.42±2.90
3	87.48±9.51	20.00±44.72	29.44±40.31	0.00±0.00	**100.00±0.00**	80.00±44.72	99.50±1.12	**100.00±0.00**
4	82.42±4.29	90.04±2.69	85.54±2.24	80.96±4.03	87.78±1.24	88.54±3.40	91.22±1.15	**91.84±0.33**
5	72.30±6.94	**87.26±9.18**	81.58±10.32	55.10±22.71	75.60±5.32	76.82±8.03	83.62±3.26	79.44±2.25
6	84.90±4.62	95.30±2.96	86.06±7.56	18.50±41.37	91.26±3.32	92.26±2.73	**96.88±1.33**	95.24±0.05
7	98.62±1.34	20.00±44.72	**100.00±0.00**	20.00±44.72	**100.00±0.00**	20.00±44.72	78.90±44.14	97.36±2.41
8	78.60±5.58	79.44±3.22	83.76±2.32	63.96±9.11	82.94±0.34	83.40±4.74	**93.12±0.28**	92.00±2.19
9	91.90±0.56	93.10±1.57	93.24±1.15	82.18±7.03	94.94±0.32	94.88±0.47	96.06±0.28	**96.32±0.11**
10	62.14±5.24	64.92±2.34	67.74±2.94	43.02±13.06	68.56±2.95	68.58±2.29	73.56±1.35	**75.52±0.11**
11	54.94±2.90	64.16±2.05	58.58±6.20	56.82±7.13	62.66±2.29	64.48±3.38	**66.08±1.51**	64.80±0.27
12	**53.94±22.39**	2.22±4.96	22.22±26.67	0.00±0.00	28.58±11.59	28.04±21.46	17.94±24.64	43.02±1.26
13	71.52±2.87	68.34±6.78	72.26±2.14	47.96±13.75	75.56±3.16	76.82±1.31	79.88±0.41	**80.30±0.27**
14	65.26±4.50	77.08±6.55	70.44±3.85	47.74±16.19	75.26±4.30	82.28±2.71	82.88±3.51	**84.48±2.85**
15	86.50±3.70	91.32±3.32	91.58±1.04	65.74±21.48	90.36±3.94	89.58±5.06	89.08±5.45	**93.34±2.14**
16	82.90±3.27	81.12±4.49	**86.34±3.17**	46.66±20.50	80.70±3.92	85.06±4.78	79.18±2.34	81.86±2.14
17	9.60±21.47	0.00±0.00	19.20±26.29	0.00±0.00	50.00±54.77	0.00±0.00	80.00±44.72	**100.00±0.00**
18	79.78±2.33	83.06±5.45	81.78±2.97	25.98±32.04	78.40±2.59	84.50±7.03	**94.36±0.96**	92.38±3.45
19	85.62±4.14	73.58±41.24	88.08±4.56	86.84±12.01	91.46±4.74	93.96±4.31	**96.80±2.28**	93.44±0.05
20	92.74±4.45	93.18±2.58	95.26±2.17	69.76±16.49	95.86±1.98	93.68±2.76	**99.30±0.54**	98.92±0.71
OA (%)	81.41±1.25	83.18±0.84	83.81±1.65	66.97±0.57	85.66±0.64	86.15±0.73	88.48±0.17	**88.65±0.03**
AA (%)	66.04±2.18	58.26±1.72	65.77±4.06	33.53±2.73	72.82±1.92	66.75±3.80	76.19±3.75	**78.91±1.02**
$k \times 100$	75.61±1.60	77.93±1.09	78.77±2.18	55.47±1.44	81.18±0.91	81.90±0.94	84.96±0.23	**85.20±0.04**

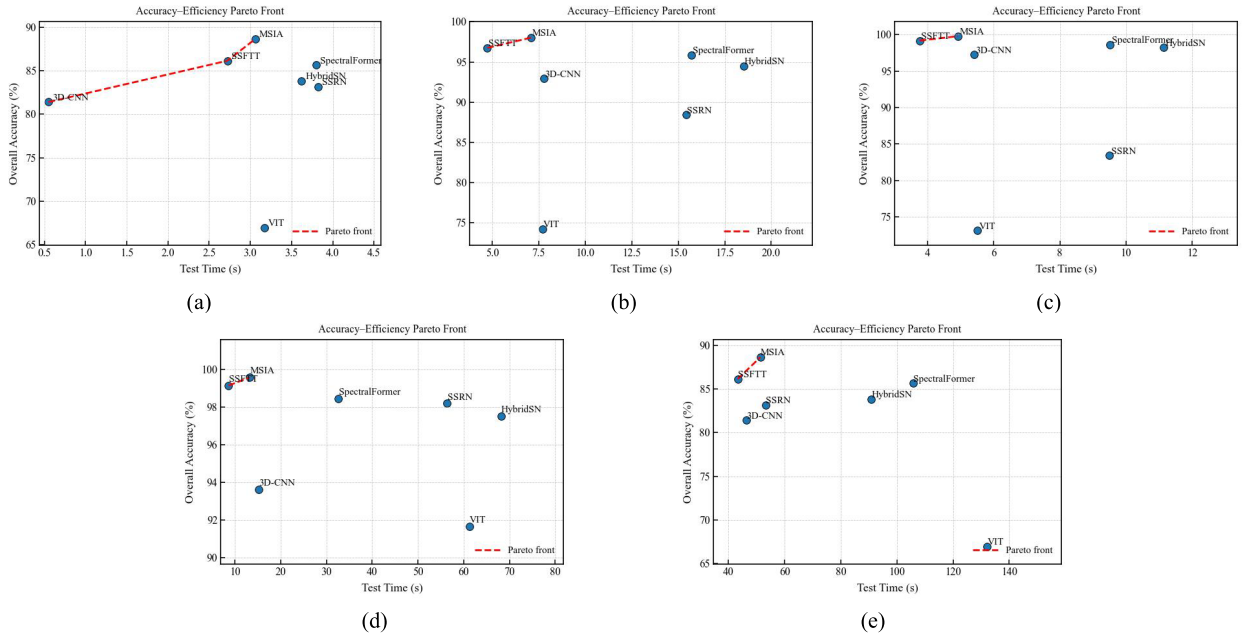

Fig. 16. Pareto front between overall accuracy and test time. (a) IP. (b) PU. (c) SA. (d) YRE. (e) HU18.

possibly due to the robust representation of the Central Region. The full MSIA model achieved the best performance across all three metrics, demonstrating that the pretraining strategy effectively improves weight initialization and mitigates overfitting under limited supervision. The detailed classification results are shown in Fig. 11.

2) *Pavia University:* The PU dataset is dominated by structured man-made objects, leading to strong performance across most models. As shown in Table VII, all models except VIT and SSRN achieved over 90% accuracy. Although VIT obtained the lowest overall accuracy (74.18%), it achieved the best performance on the Shadows (Class 9), which had the fewest training samples. This highlights a key limitation of single-scale global modeling: when training samples are scarce, models that overly emphasize global consistency may be biased toward simpler spectral patterns, compromising their ability to distinguish more complex categories. The performance gap between MSIA and MSIA-WP is relatively small on this dataset, indicating that pretraining does not significantly improve the performance

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TABLE XI
MODEL COMPLEXITY COMPARISON

Method	Params(M)	Flops(MMac)	Training Time(s)	Test Time(s)
IP				
3D-CNN	1.91	27.82	48.32	0.55
SSRN	0.36	46.19	97.70	3.82
HybridSN	5.12	136.64	123.04	3.62
VIT	2.04	1.42	50.71	3.17
SpectralFormer	0.52	41.49	87.26	3.80
SSFTT	0.15	5.70	27.29	2.72
MSIA	0.82	18.28	42.63	3.06
PU				
3D-CNN	1.91	27.82	108.85	7.79
SSRN	0.36	46.19	205.43	15.42
HybridSN	5.12	136.64	95.71	18.51
VIT	1.44	1.11	99.36	7.70
SpectralFormer	0.52	41.49	200.39	15.71
SSFTT	0.15	5.70	67.13	4.73
MSIA	0.22	16.87	101.53	7.09
SA				
3D-CNN	1.91	27.82	138.92	5.40
SSRN	0.36	46.41	283.36	9.50
HybridSN	5.12	136.64	368.96	11.14
VIT	2.10	1.43	157.98	5.51
SpectralFormer	0.52	41.49	253.23	9.52
SSFTT	0.15	5.70	81.20	3.76
MSIA	0.22	16.87	130.42	4.92
YRE				
3D-CNN	1.91	27.82	202.53	15.23
SSRN	0.36	46.02	427.63	56.38
HybridSN	5.12	136.64	540.36	68.17
VIT	2.55	1.67	273.53	61.30
SpectralFormer	0.52	41.49	365.12	32.62
SSFTT	0.15	5.70	116.40	8.63
MSIA	0.36	7.42	182.08	13.23
HU18				
3D-CNN	1.91	27.82	96.46	46.32
SSRN	0.36	34.94	98.64	53.30
HybridSN	5.12	136.64	178.87	90.83
VIT	1.11	0.95	145.15	132.12
SpectralFormer	0.52	41.49	145.89	105.70
SSFTT	0.15	5.70	82.63	43.28
MSIA	0.22	16.87	98.36	51.36

upper bound. However, it does enhance training stability across repeated experiments. Classification maps are shown in Fig. 12.

3) *SAs Dataset:* Table VIII summarizes the classification results on the SAs dataset. Apart from a further decline in the performance of SSRN and ViT, other models exhibit similar performance trends to those observed on the PU dataset. Comparing these results across datasets reveals that full-spectrum models tend to perform worse as spectral resolution increases. The performance gap between MSIA and MSIA-WP is negligible on this dataset, indicating that the benefit of pretraining diminishes when sufficient labeled samples are available. The classification results are visualized in Fig. 13.
4) *Yellow River Estuary Coastal Wetland Dataset:* Table IX presents the classification results on the YRE dataset. Despite the limited number of training samples, all models achieved effective training due to the high quality of manually labeled samples. SSRN, in particular, recovered its expected performance with an overall accuracy of 98.19%. Among all methods, MSIA consistently achieved the best performance. The corresponding classification maps are shown in Fig. 14.
5) *Houston2018:* The HU18 dataset represents a fine-grained urban scene characterized by substantial variation in object scales, which challenges the model's ability to capture multiscale information. The primary difficulty of this dataset lies in distinguishing small-scale scenes that resemble large-scale ones, thereby testing the model's discriminative capability beyond spatial inertia. Under the proposed multiscale structural partitioning of MSIA, the model achieves leading accuracy not only in the commercial class (Class 9) but also in challenging, visually similar categories such as residential (Class 8) and cars (Class 18). A similar trend is observed for Paved Parking Lot (Class 16) and Gravel Parking Lot (Class 17), where MSIA also attains superior performance. The corresponding classification maps are shown in Fig. 15.

E. Model Complexity Analysis

To assess MSIA's efficiency and complexity, we compared all methods across five datasets in terms of parameters, FLOPs, and average training/testing time (three runs), with consistent settings as in the Experimental Setup, and the detailed results are summarized in Table XI. Overall, MSIA demonstrates promising performance in terms of model compactness and computational efficiency. Although its parameter count on the IP dataset is slightly higher due to the larger number of retained principal components, it remains among the lowest on the other datasets, second only to SSFTT. While MSIA exhibits slightly higher FLOPs than other Transformer-based methods due to its architectural complexity, it maintains the second fastest training time.

Fig. 16 shows the accuracy-efficiency Pareto front. Each point represents a model with its overall accuracy and test time. The red dashed line highlights the Pareto frontier, illustrating the optimal tradeoff between accuracy and efficiency. MSIA lies near the frontier, demonstrating competitive efficiency while maintaining state-of-the-art accuracy.

F. Ablation Experiment

1) *Ablation Study of the Proposed:* To assess the contribution of each submodule within MSIA to the overall performance, we conducted ablation experiments on all five datasets. Specifically, we evaluated three model variants: using only the center region transformer (CRT), using only the CAB, and the full MSIA model. The results are presented in Table XII. The findings indicate that the CAB plays a pivotal role, demonstrating strong effectiveness in multiscale feature integration. While the CRT module alone yields relatively lower accuracy, its inclusion enhances the representation of center-region features when combined with CAB. Overall, the fusion

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TABLE XII
PERFORMANCE CONTRIBUTION OF EACH MODULE IN THE MSIA FRAMEWORK

CRT	CAB	IP			PU			SA			YRE			HU18		
		OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA	OA	AA	KAPPA
√		82.83	64.58	80.28	86.98	75.98	82.43	88.33	85.89	86.97	71.43	43.68	63.65	68.71	37.35	58.71
	√	95.11	74.83	94.41	97.64	96.15	96.87	98.42	99.08	98.21	97.81	96.23	97.34	86.84	69.35	82.86
√	√	97.32	93.61	97.21	98.16	96.49	97.56	99.77	99.68	99.75	99.70	99.51	99.64	88.69	77.80	85.24

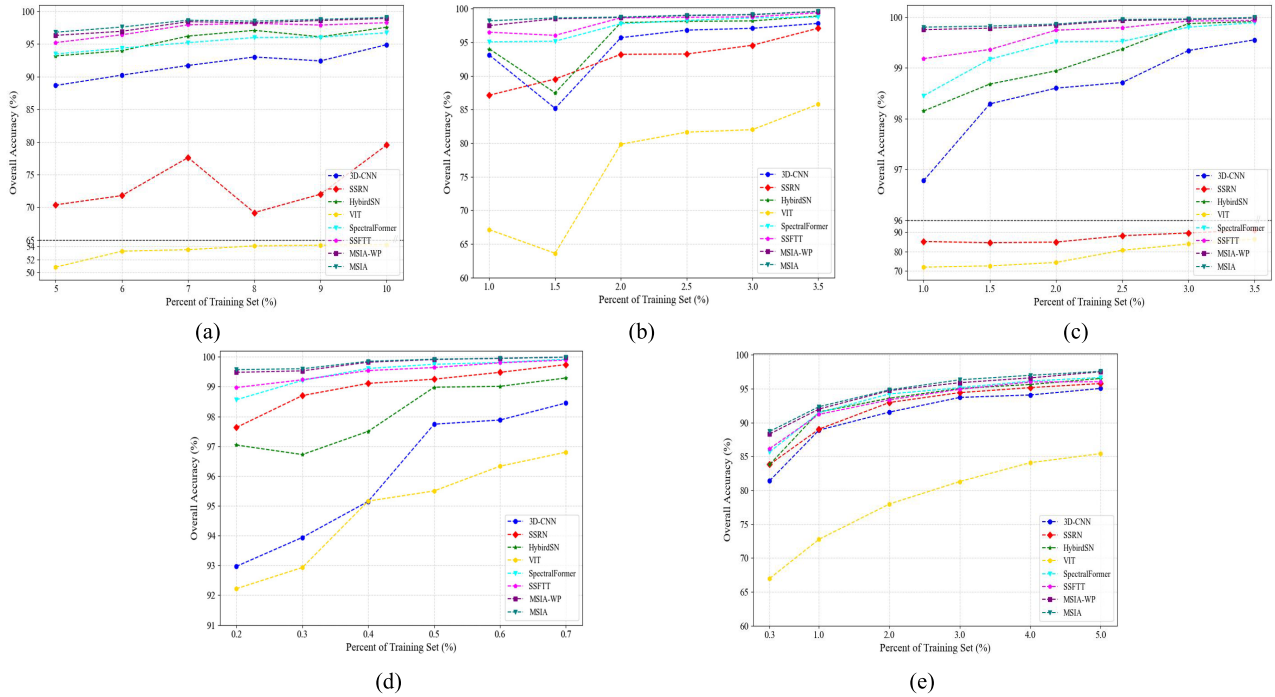

Fig. 17. Classification performance under different training samples. (a) IP. (b) PU. (c) SA. (d) YRE. (e) HU18.

of both modules leads to the best classification performance, confirming their complementary strengths.

2) *Ablation Study of the Percentage of Training Samples:* We further evaluated the robustness of each method under different supervision levels by comparing their overall accuracy across various training sample ratios: 5%–10% for IP, 1%–3.5% for PU and SA, 0.2%–0.7% for YRE, and 0.3%–5% for HU18. As shown in Fig. 17, although some methods fluctuate at certain sampling rates, all models exhibit a consistent increase in OA as the number of labeled samples grows.

3) *Effect of the Concentric Scaling Strategy:* To assess whether the proposed concentric partition configuration (W, $2W+1$, $4W+3$) is optimal or heuristic, we conducted an ablation study comparing different region scaling schemes. Specifically, experiments were conducted on the IP and HU18 datasets using center region sizes of 3 and 5 to compare two scaling strategies: Extended scaling (W, $2W+1$, $6W+5$) and Linear scaling (W, $W+2$, $W+6$).
As shown in Table XIII, Extended scaling captures broader contextual information but tends to introduce redundant background pixels, slightly reducing accuracy and increasing inference time. In contrast, linear scaling provides faster inference but captures less contextual information, leading to weaker spatial–spectral representations. These results confirm that the proposed scaling achieves a more balanced tradeoff between detail preservation and contextual understanding.

4) *Ablation Study of the Query-Key Asymmetry in CAB:* To validate the rationale of the asymmetric design in the CAB module, where near regions serve as query and relevant regions provide key-value pairs, we compared three configurations: 1) proposed CAB (near to query, relevant to key and value); 2) inverse-asymmetric CAB (relevant to query, near to key and value); and 3) equal self-attention, which applies a Transformer directly to the entire relevant region. Experiments were conducted on the IP and HU18 datasets.
As shown in Table XIV, the inverse-asymmetric design results in a notable accuracy drop compared with the proposed CAB, indicating that using noisy global features as queries weakens spatial discrimination. The equal self-attention further decreases accuracy and increases inference time due to redundant interactions within the large relevant region.

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TABLE XIII
EFFECT OF DIFFERENT SCALING STRATEGIES ON CLASSIFICATION PERFORMANCE

Datasets	center region size	Metrics	Linear scaling	Proposed scaling	Extended scaling
IP	3	OA(%)	96.22	97.32	97.02
		Flops(MMac)	1.1	18.28	67.52
		Test time(s)	2.30	3.06	4.66
	5	OA(%)	96.57	97.14	96.66
		Flops(MMac)	1.26	72.15	165.61
		Test time(s)	2.85	6.44	8.85
HU18	3	OA(%)	86.37	86.83	88.11
		Flops(MMac)	0.24	0.36	15.23
		Test time(s)	21.84	36.42	48.75
	5	OA(%)	87.04	88.69	87.59
		Flops(MMac)	0.28	16.87	41.45
		Test time(s)	24.29	51.36	81.12

TABLE XIV
COMPARISON OF DIFFERENT QUERY-KEY CONFIGURATIONS IN THE CAB

Datasets	Metrics	Proposed CAB	inverse-asymmetric CAB	equal self-attention
IP	OA(%)	97.32	95.52	92.55
	Flops(MMac)	18.28	16.28	22.88
	Test time(s)	3.06	2.97	3.15
HU18	OA(%)	88.69	79.69	76.53
	Flops(MMac)	16.87	15.87	9.87
	Test time(s)	51.36	46.80	75.95

TABLE XV
EFFECT OF DIFFERENT ENCODERS FOR THE CENTER REGION

Datasets	Metrics	Transformer encoder	Stacked Conv Encoder	Conv + Attention
IP	OA(%)	97.23	93.94	94.82
	Flops(MMac)	18.28	17.60	17.61
	Test time(s)	3.06	1.93	2.28
HU18	OA(%)	88.32	85.26	85.46
	Flops(MMac)	16.87	17.60	17.61
	Test time(s)	51.36	50.68	56.24

5) *Effect of Different Encoders for the Center Region:* To evaluate whether a full Transformer encoder is necessary for the center region, we compared it with two lightweight alternatives: 1) a stacked convolutional encoder composed of three 3×3 convolution kernels and 2) a Conv + Attention module, which integrates a CBAM attention block in the convolutional stack. Experiments were conducted on the IP and HU18 datasets.

As shown in Table XV, the Transformer encoder introduces only a negligible increase in computational cost compared with the convolutional encoder, yet achieves 1.8%–3.0% higher accuracy than its convolution-based variants. The Conv + Attention variant provides only a slight improvement over pure convolution, indicating that simple channel–spatial recalibration cannot fully replace the dynamic token interactions modeled by self-attention. These results suggest that even in small regions, the Transformer encoder effectively captures subtle pixel dependencies and yields more discriminative representations.

6) *Effect of Attention Heads and Positional Encoding Attention Heads in the CRT:* Further ablation studies on the CRT were conducted to examine two key factors: the number of attention heads in the MHSA mechanism. These experiments were performed on the IP and PU datasets, and the use of positional encoding was considered. These experiments were performed on the IP and PU datasets.

   1) As shown in Table XVI shows that using four heads yields the best overall performance on both datasets. Too few heads limit the model's ability to capture diverse spatial–spectral dependencies, while too many introduce redundancy without further gains.
   2) As shown in Table XVII, removing positional embeddings caused a clear decline in both overall and average accuracy on the IP and PU datasets, confirming that explicit positional information is crucial for maintaining spatial awareness and distinguishing spectrally similar but spatially different pixels.

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TABLE XVI
EFFECT OF NUMBER OF ATTENTION HEADS ON CLASSIFICATION PERFORMANCE

Number of attention heads	IP			PU		
	OA	AA	KAPPA	OA	AA	KAPPA
2	96.66	93.30	96.18	98.08	96.43	97.45
4	97.32	93.61	97.21	98.16	96.49	97.56
8	97.05	92.49	96.63	97.92	96.09	97.24

TABLE XVII
EFFECT OF POSITIONAL ENCODING ON CLASSIFICATION PERFORMANCE

Frame work	IP			PU		
	OA	AA	KAPPA	OA	AA	KAPPA
With positional embeddings	97.32	93.61	97.21	98.16	96.49	97.56
Without positional embeddings	95.93	91.11	95.34	96.85	94.17	95.81

V. DISCUSSION

The superior performance of MSIA across datasets validates the effectiveness of the hierarchical region partitioning and central-pixel enhancement strategies. In scenes with large variations in object scales, visual comparison with ground truth shows that the proposed model produces fewer noise artifacts in large homogeneous areas and generates clearer boundaries at class transitions. For example, in the YRE dataset, MSIA accurately captures the detailed transition between Bare soil (Class 2) and Suaeda salsa (Class 4). In large-scale scenes containing small objects, the model also preserves the integrity and contour precision of fine targets. In the HU18 dataset, for instance, cars (Class 18) maintain clear boundaries and shapes between residential (Class 8) and major thoroughfares (Class 13). This demonstrates that the CAB module enhances multiscale contextual consistency, effectively suppressing noise and false responses in extensive homogeneous regions. The central-region Transformer further strengthens focus on key pixels, yielding sharper contours and finer details for small objects.

In terms of efficiency, MSIA achieves moderate performance—it requires slightly longer inference time than SSFTT but remains faster than convolution-based models.

Comparative experiments also reveal class-specific effects of pretraining. For classes with complex spectra or high sensitivity to background noise—such as Corn-notill (IP Class 2), Bricks (PU Class 8), and Tamarix (YRE Class 3)—pretraining notably improves accuracy by over 2%. For most other classes, the gains are minor (less than 1%) or slightly negative. However, pretraining consistently enhances model stability, as evidenced by the reduced standard deviation compared with randomly initialized models. Although its benefit becomes less pronounced with abundant labeled samples, it still reduces performance variance, indicating better initialization and convergence behavior.

Although the proposed MSIA framework achieves competitive performance on multiple hyperspectral datasets, it still has several limitations. First, it relies on manually defined concentric regions, whose scale settings may not be optimal for all scenarios. Second, the model's complex structure introduces additional computational overhead, which may restrict its deployment in edge or real-time applications. Future work will explore adaptive region partitioning strategies, lightweight attention mechanisms, and large-scale cross-scene evaluations to further enhance efficiency and robustness.

VI. CONCLUSION

To overcome the limitations of single-scale modeling and unclear central-pixel representation in HSI classification, this article presents an MSIA. The model hierarchically partitions each input into center, near, and relevant regions, employs a Transformer encoder to enhance central features, and introduces a CAB for selective cross-scale interaction between local and global contexts. Furthermore, a pretraining strategy is incorporated to leverage abundant unlabeled data for learning robust feature representations. Experiments on five benchmark datasets indicate that MISA performs favorably compared with existing advanced methods.

While these results confirm the effectiveness of the proposed framework, MISA still relies on fixed region settings and involves a relatively high computational cost. Future work will focus on adaptive region partitioning and lightweight model designs to further improve flexibility and deployment efficiency.

REFERENCES

[1] K. Tan, F. Wu, Q. Du, P. Du, and Y. Chen, "A parallel Gaussian–Bernoulli restricted Boltzmann machine for mining area classification with hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 2, pp. 627–636, Feb.2019, doi: 10.1109/JSTARS.2019.2892975.

[2] G. J. Lakshmi, V. Bodapati, S. S. Baji, and T. S. Panuganti, "Water quality monitoring using remote-sensing," in *Proc. Int. Conf. Comput. Intelli-gence Sustain. Eng. Solutions (CISES)*, 2023, pp. 624–628, doi: 10.1109/cises58720.2023.10183395.

[3] Z. Xue, P. Du, J. Li, and H. Su, "Sparse graph regularization for robust crop mapping using hyperspectral remotely sensed imagery with very few in situ data," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 1–15, Feb.2017, doi: 10.1016/j.isprsjprs.2016.12.003.

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[4] T. C. W. Mok and A. C. S. Chung, "Fast symmetric diffeomorphic image registration with convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4643–4652, doi: 10.1109/CVPR42600.2020.00470.

[5] J. Zhang, R. Zhao, Z. Shi, N. Zhang, and X. Zhu, "Bayesian constrained energy minimization for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8359–8372, 2021.

[6] C. Tao, Y. Tang, C. Fan, and Z. Zou, "Hyperspectral imagery classification based on rotation-invariant spectral–spatial feature," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 980–984, May2014.

[7] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Phil. Trans. Roy. Soc. A., Math., Phys. Eng. Sci.*, vol. 374, no. 2065, Apr.2016, Art. no. 20150202.

[8] A. M. Martinez and A. C. Kak, "PCA versus LDA," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 23, no. 2, pp. 228–233, Feb.2001.

[9] L. M. Bruce, C. H. Koger, and J. Li, "Dimensionality reduction of hyperspectral data using discrete wavelet transform feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2331–2338, Jan.2002.

[10] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based *k*-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov.2010, doi: 10.1109/TGRS.2010.2055876.

[11] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar.2005, doi: 10.1109/TGRS.2004.842481.

[12] Y. Xu et al., "MambaHSISR: Mamba hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511216.

[13] Y. Xu et al., "A comb concatenation diffusion model for hyperspectral image super-resolution," *Eng. Appl. Artif. Intell.*, vol. 160, Nov.2025, Art. no. 111985.

[14] Y. Fu, Z. Liu, and J. Lyu, "Reason and discovery: A new paradigm for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5586–5599, Jul.2025.

[15] H. Wang, S. Sun, and P. Ren, "Underwater color disparities: Cues for enhancing underwater images toward natural color consistencies," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 738–753, Feb.2024.

[16] X. Tao, B. Koirala, A. Plaza, and P. Scheunders, "A new dual-feature fusion network for enhanced hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540113.

[17] Y. Zhao, M. Jia, G. Sun, and A. Zhang, "PAMSNet: A point annotation-driven multi-source network for remote sensing semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 229, pp. 1–16, Nov.2025.

[18] A. Mughees and L. Tao, "Multiple deep-belief-network-based spectral–spatial classification of hyperspectral images," *Tsinghua Sci. Technol.*, vol. 24, no. 2, pp. 183–194, Apr.2019.

[19] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun.2014.

[20] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul.2017.

[21] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug.2018, doi: 10.1109/TGRS.2018.2818945.

[22] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun.2018, doi: 10.1109/TGRS.2018.2794326.

[23] H. Ge et al., "Pyramidal multiscale convolutional network with polarized self-attention for pixel-wise hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504018, doi: 10.1109/TGRS.2023.3244805.

[24] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Nov.2015.

[25] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct.2016.

[26] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec.2017.

[27] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb.2018.

[28] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb.2020, doi: 10.1109/LGRS.2019.2918719.

[29] D. Yu, Q. Li, X. Wang, Z. Zhang, Y. Qian, and C. Xu, "DSTrans: Dual-stream transformer for hyperspectral image restoration," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Wai-koloa, HI, USA, Jan. 2023, pp. 3728–3738, doi: 10.1109/WACV56688.2023.00373.

[30] F. Deng, W. Luo, Y. Ni, X. Wang, Y. Wang, and G. Zhang, "UMiT-Net: A U-shaped mix-transformer network for extracting precise roads using remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5801513, doi: 10.1109/TGRS.2023.3281132.

[31] J. Yang, B. Du, and L. Zhang, "From center to surrounding: An interactive learning framework for hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 145–166, Mar.2023.

[32] A. Dosovitskiy, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent.*, May 2021, pp. 1–21.

[33] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

[34] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[35] Y. Liu, J. Hu, X. Kang, J. Luo, and S. Fan, "Interactformer: Interactive transformer and CNN for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531715, doi: 10.1109/TGRS.2022.3183468.

[36] Y. Wang, X. Yu, X. Wen, X. Li, H. Dong, and S. Zang, "Learning a 3-D-CNN and convolution transformers for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[37] X. Tang, Y. Zou, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Multiple information collaborative fusion network for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525416.

[38] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3144158.

[39] X. Hu, T. Li, T. Zhou, Y. Liu, and Y. Peng, "Contrastive learning based on transformer for hyperspectral image classification," *Appl. Sci.*, vol. 11, no. 18, Sep.2021, Art. no. 8670.

[40] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6008205.

[41] S. Hou, H. Shi, X. Cao, X. Zhang, and L. Jiao, "Hyperspectral imagery classification based on contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521213.

[42] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[43] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual repre-sentations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, Sep. 2020, pp. 1597–1607.

[44] J.-B. Grill, "Bootstrap your own latent—A new ap-proach to self-supervised learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst., Jul.*, 2020, pp. 21271–21284.

[45] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15750–15758.

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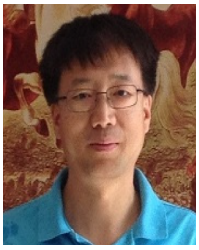

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